COMPUTER NETWORKS

Textbook

- **Behrouz A. Forouzan**, **Data Communications and Networking, 5th Edition**, Tata McGraw-Hill,2013

Reference Books:

1. Larry L. Peterson and Bruce S. Davie: **Computer Networks – A Systems Approach, 4th Edition**, Elsevier, 2019.

2. Nader F. Mir: Computer and Communication Networks, 2nd Edition, Pearson Education, 2015.

3. William Stallings, Data and Computer Communication 10th Edition, Pearson Education, Inc., 2014

What is Data?

What is Communication?

What is Data?
Data are raw facts, figures, symbols, or observations that by themselves may not carry meaning.

Data → Raw facts

When data are processed, organized, or interpreted, they become information.

Examples of data:
Numbers: 85, 120, 3.14
Text: Delhi, A, Student
Symbols: @, #, %
Images, audio clips, sensor readings

Data

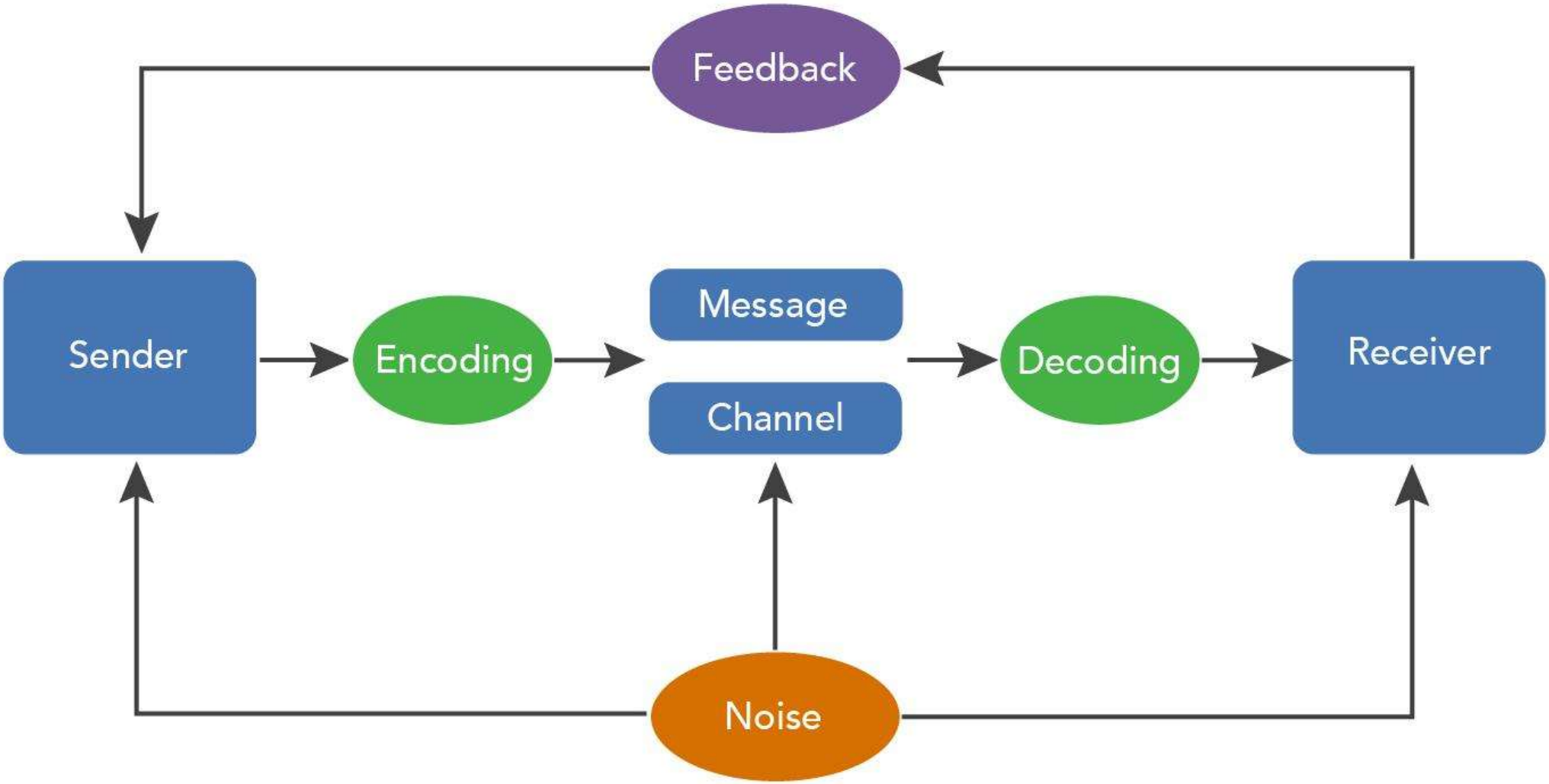
- A collection of facts or statistics
- Unorganized
- Without Context
- Can be quantitative (numerical) or qualitative (descriptive)
- Examples: number of website visitors, individual customer survey responses, product price

Information

- The result of analyzing and interpreting data
- Has context
- Can be used to help make decisions
- Examples: website traffic changes, customer sentiment based on survey results, product price comparison

What is Communication?
Communication is the process of transmitting data or information from a sender to a receiver through a medium (channel), so that the message is understood.

Elements of communication:
Sender – who sends the message
Message – the data/information being sent
Medium/Channel – path (wire, air, internet, etc.)
Receiver – who receives the message
Feedback – response from the receiver


Feedback
Sender
Encoding
Message
Channel
Decoding
Receiver
Noise

Data Communications

- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- **Four fundamental characteristics** of Data Communication are:

1. Delivery: The system must deliver data to the correct destination.

2. Accuracy: The system must deliver the data accurately.

3. Timeliness:

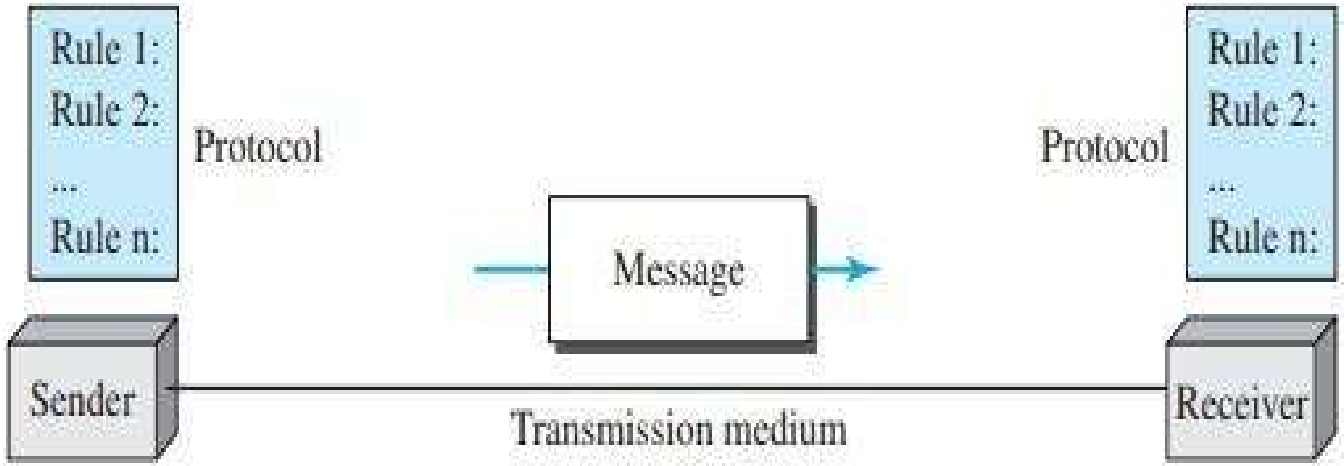
- The system must deliver data in a timely manner.
- Data delivered **late are useless.**
- In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.

4. Jitter:

- Jitter refers to the variation in the packet arrival time.
- **For example, let us assume that video packets are sent every 30 ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result**.

Data Communications- Components

Figure 1.1 *Five components of data communication*



Data Communications- Components(Contd...)

1. **Message:** The message is the information (data) to be communicated. Popular forms of information include text, numbers, audio, and video.
2. **Sender:** The sender is the device that sends the data message. It can be a **computer, workstation etc**
3. **Receiver:** The receiver is the device that receives the message. It can be a **computer, workstation etc.**
4. **Transmission medium:** The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include **twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves**.
5. **Protocol:** A protocol is a **set of rules** that govern data communications. Without a protocol, **two devices may be connected but not communicating.**

What are the different types of data and how is data represented ?

There are 5 main types of data:

1.Numeric Data – Numbers used for calculations

Example: 10, 25.6, −7

2.Text (Alphanumeric) Data – Letters, digits, symbols

Example: ABC123, Delhi@2025

3.Image Data – Visual information stored as pixels

Example: JPG, PNG

4.Audio Data – Sound data stored digitally

Example: MP3, WAV

5.Video Data – Moving images with audio

Example: MP4, AVI

Data Communications- Data Representation

1. Text:

- Text is represented as a **bit pattern**, a sequence of bits (0s or 1s)
- Different sets of bit patterns have been designed to represent text symbols. Each set is called a **code,** and the process of representing symbols is called **coding.**

Character	ASCII Code	Binary (Bit Pattern)
H	72	`01001000`
e	101	`01100101`
l	108	`01101100`
l	108	`01101100`
o	111	`01101111`

2. Numbers:

- Numbers are also represented by **bit patterns** and the number is directly converted to a **binary number**.

Decimal: 13

Binary (Bit Pattern): `00001101`

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Binary (Bit Pattern): `00001101`

Data Communications- Data Representation

3. Images:

- Images are also represented by **bit patterns**
- In its simplest form, an image is composed of a **matrix of pixels** (picture elements), where each pixel is a **small dot.** The size of the pixel depends on the **resolution**
- After an image is divided into pixels, each pixel is assigned a bit pattern.
- If the image made only on **black and-white** dots (0 or 1)
- In the case of **Gray scale**

  - black pixel - 00
  - dark gray pixel- 01
  - light gray pixel- 10
  - white pixel -11

Data Communications- Data Representation

- **RGB :** Each color is made of a combination of three primary colors: **red, green, and blue**
- **YCM :** color is made of a combination of three other primary colors: **yellow, cyan, and magenta**

4. Video:

- Video refers to the recording or broadcasting of a picture or movie
- It may be a **continuous entity** or **combination of images, arranged to convey the idea of motion**

5. Audio :

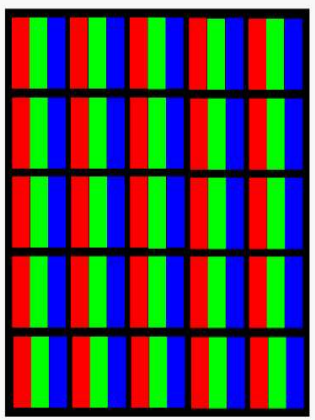
- Audio refers to the recording or broadcasting of sound or music.
- It is continuous, not discrete

Data Communications- Data Representation

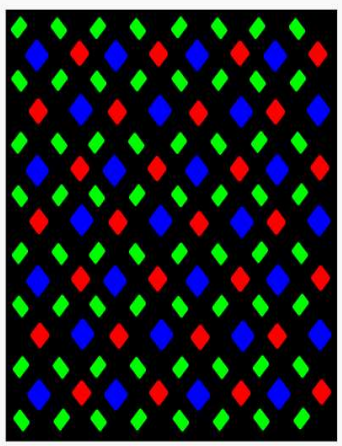
- **Basic form of representation:** Binary (0 and 1)
- **Text representation:** ASCII, Unicode
- **Image:** Pixels
- **Audio:** Sampling
- **Video:** Frames + audio

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	00	Null	32	20	Space	64	40	@	96	60	`
1	01	Start of heading	33	21	!	65	41	A	97	61	a
2	02	Start of text	34	22	"	66	42	B	98	62	b
3	03	End of text	35	23	#	67	43	C	99	63	c
4	04	End of transmit	36	24	$	68	44	D	100	64	d
5	05	Enquiry	37	25	%	69	45	E	101	65	e
6	06	Acknowledge	38	26	&	70	46	F	102	66	f
7	07	Audible bell	39	27	'	71	47	G	103	67	g
8	08	Backspace	40	28	(	72	48	H	104	68	h
9	09	Horizontal tab	41	29	)	73	49	I	105	69	i
10	0A	Line feed	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage return	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	47	2F	/	79	4F	O	111	6F	o
16	10	Data link escape	48	30	0	80	50	P	112	70	p
17	11	Device control 1	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	50	32	2	82	52	R	114	72	r
19	13	Device control 3	51	33	3	83	53	S	115	73	s
20	14	Device control 4	52	34	4	84	54	T	116	74	t
21	15	Neg. acknowledge	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	54	36	6	86	56	V	118	76	v
23	17	End trans. block	55	37	7	87	57	W	119	77	w
24	18	Cancel	56	38	8	88	58	X	120	78	x
25	19	End of medium	57	39	9	89	59	Y	121	79	y
26	1A	Substitution	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	60	3C	<	92	5C	\	124	7C	|
29	1D	Group separator	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	63	3F	?	95	5F	_	127	7F	□

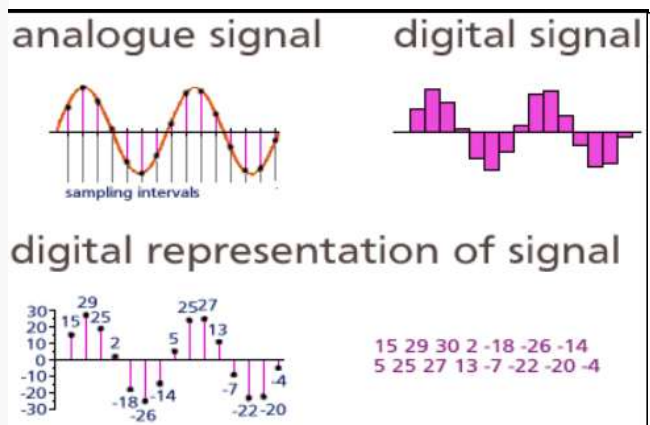
b_7 b_6 b_5 →					0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
Bits b_4 ↓	b_3 ↓	b_2 ↓	b_1 ↓	Column → / Row ↓	0	1	2	3	4	5	6	7
0	0	0	0	0	NUL	DLE	SP	0	@	P	`	p
0	0	0	1	1	SOH	DC1	!	1	A	Q	a	q
0	0	1	0	2	STX	DC2	"	2	B	R	b	r
0	0	1	1	3	ETX	DC3	#	3	C	S	c	s
0	1	0	0	4	EOT	DC4	$	4	D	T	d	t
0	1	0	1	5	ENQ	NAK	%	5	E	U	e	u
0	1	1	0	6	ACK	SYN	&	6	F	V	f	v
0	1	1	1	7	BEL	ETB	'	7	G	W	g	w
1	0	0	0	8	BS	CAN	(	8	H	X	h	x
1	0	0	1	9	HT	EM	)	9	I	Y	i	y
1	0	1	0	10	LF	SUB	*	:	J	Z	j	z
1	0	1	1	11	VT	ESC	+	;	K	[	k	{
1	1	0	0	12	FF	FS	,	<	L	\	l	|
1	1	0	1	13	CR	GS	—	=	M	]	m	}
1	1	1	0	14	SO	RS	.	>	N	^	n	~
1	1	1	1	15	SI	US	/	?	O	_	o	DEL

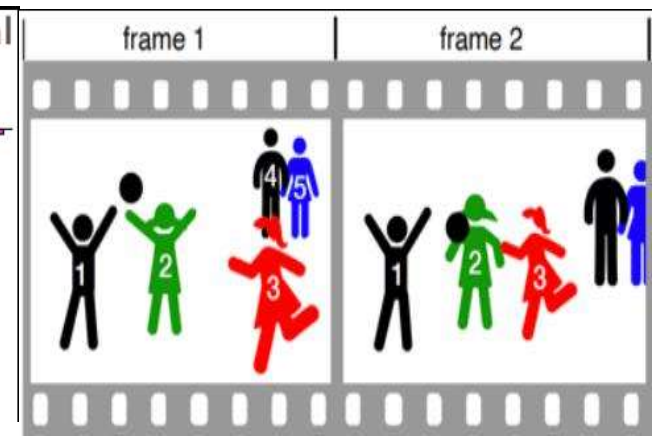


5 x 5 pixel with RGB subpixels



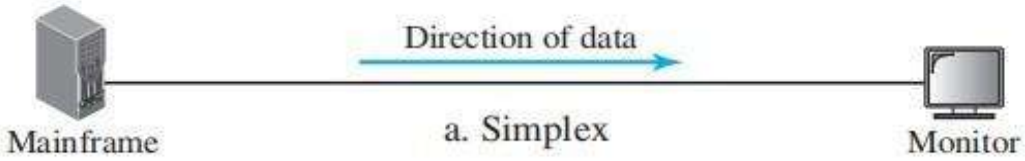
Subpixel layout of an iPhone





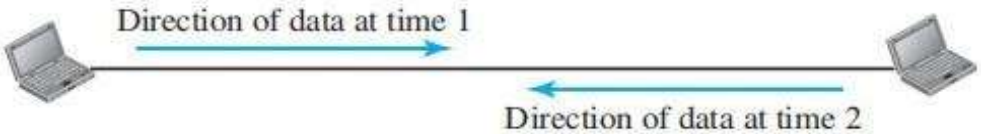
Data Communications-Data Flow

- **Simplex :**



a. Simplex

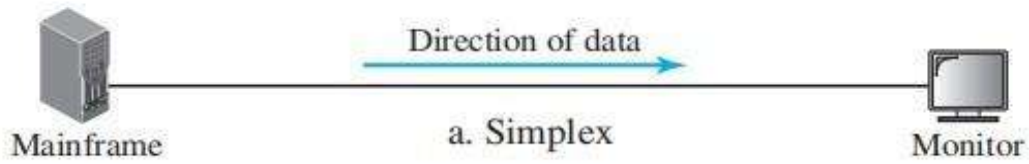
- **Half-duplex**



b. Half-duplex

Data Communications-Data Flow

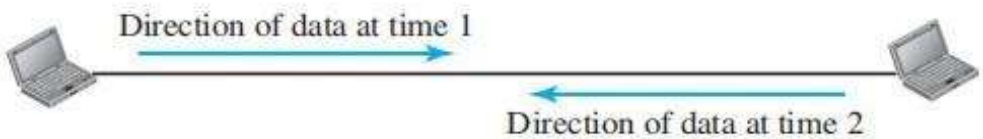
- **Simplex :**



a. Simplex

Examples:
•Keyboard → Computer
•Television broadcast
•Radio transmission

- **Half-duplex**

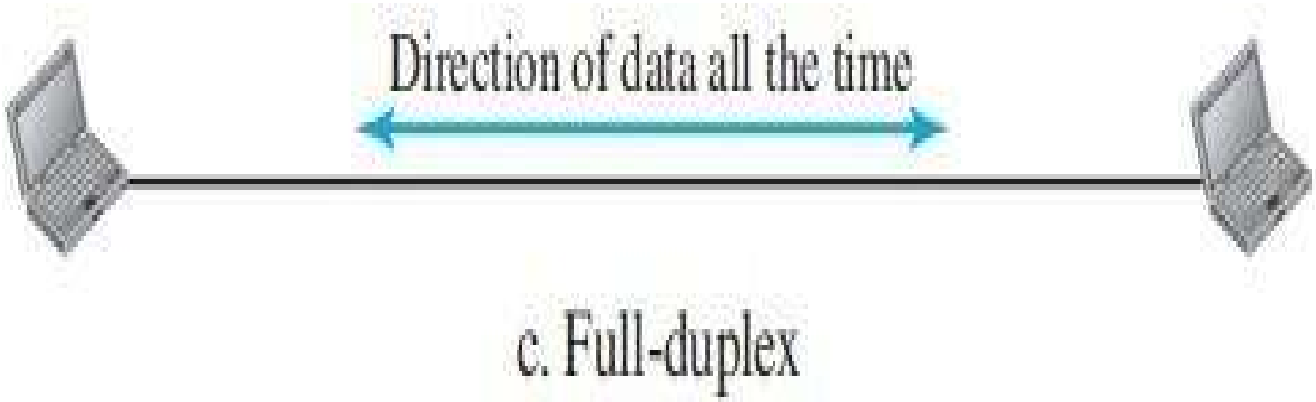


b. Half-duplex

Examples:
•Walkie-talkie
•CB radio
•Push-to-talk communication systems

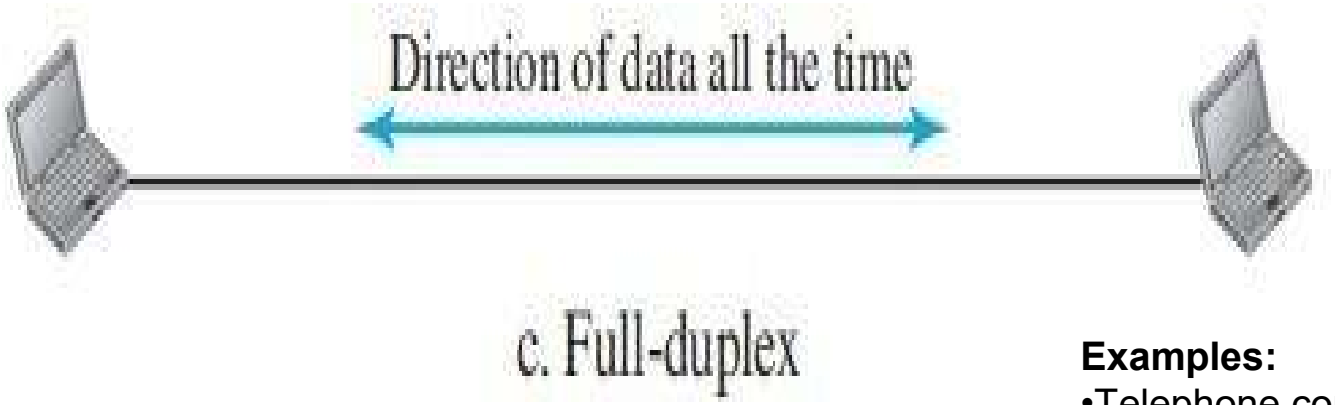
Data Communications-Data Flow

- **Full duplex**



Data Communications-Data Flow

- **Full duplex**



Examples:
•Telephone conversation
•Mobile phone calls
•Video conferencing

Comparison Chart: Simplex vs Half-Duplex vs Full-Duplex

Attribute	Simplex	Half-Duplex	Full-Duplex
Direction of Data Flow	One-way only	Two-way (one at a time)	Two-way (simultaneous)
Simultaneous Transmission	Not possible	Not possible	Possible
Feedback from Receiver	No	Yes (but delayed)	Yes (instant)
Channel Utilization	Low	Medium	High
Complexity	Very simple	Moderate	Complex
Efficiency	Low	Medium	High
Examples	TV, Radio	Walkie-talkie	Telephone, Video call

NETWORKS

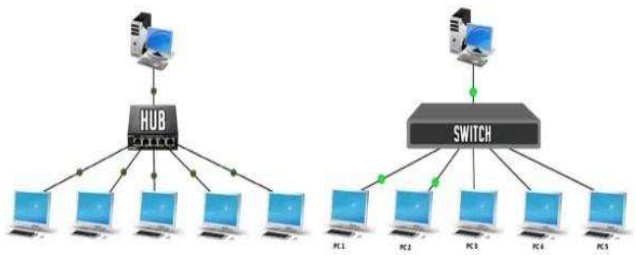
- A network is the **interconnection** of a set of **devices** capable of communication.

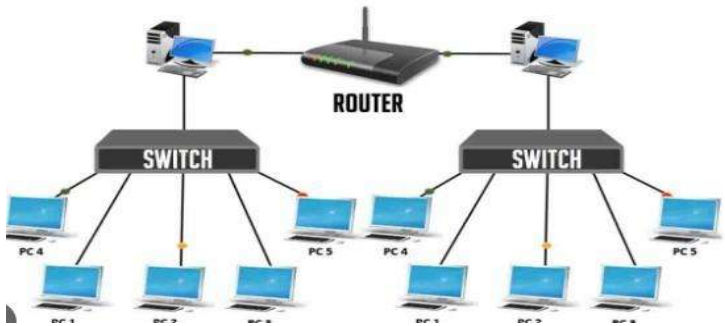
- **Device** can be a **host or node :** large **computer, desktop, laptop, workstation, cellular phone, or security system etc..**

NETWORKS

- **Connecting device :**

Router- which connects the network to other networks
Switch- which connects devices together
Modem (modulator-demodulator)- which changes the form of data





- Devices in a network are connected using wired or wireless transmission media such as **cable or air**

NETWORKS - Network Criteria

- A Network must be able to meet a certain number of criteria

1. **Performance**
   - It measures based on **transit time** and response time
   - Performance is often evaluated by two networking metrics: **throughput and delay**
     - ***Throughput:Number of bits transmitted in per second***
     - ***Delay: time taken to transmit data***

2. **Reliability**
   - Reliable networking ensures that data is transmitted accurately, in order, and without errors between devices or systems.

3. **Security**
   - Protecting data from unauthorized access, protecting data from damage, development.

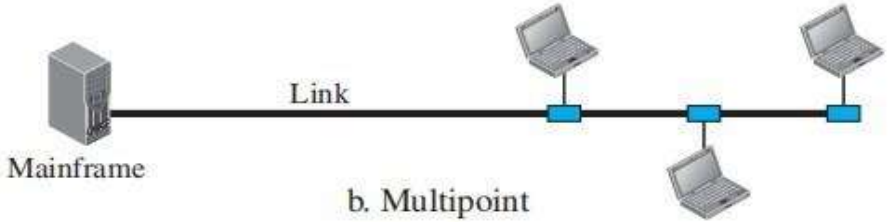
NETWORKS -Types of Connection

Types of connection describe the way devices communicate with each other over a network

- **Point-to-Point**

a. Point-to-point

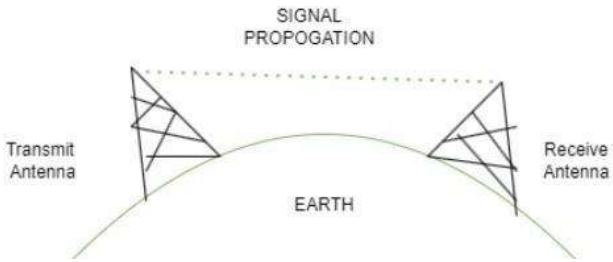
- **Multipoint or Multidrop**



b. Multipoint

NETWORKS -Types of Connection

- **Point-to-Point**
  - ***Dedicated link between two devices***
  - ***The entire capacity of the channel is reserved***
  - ***Eg., Microwave link, TV remote control***



- **Multipoint or Multidrop**
  - ***More than two devices share a single link***
  - ***Capacity of the channel is either***
    - ***Spatially shared:*** ***Devices can use the link simultaneously***
    - ***Timeshare:*** ***Users must take turns***

NETWORKS -Physical Topology

- **Physical Topology :** Physical arrangement of devices (nodes) in a network and how they are interconnected.
- Connecting the devices **geometrically**.

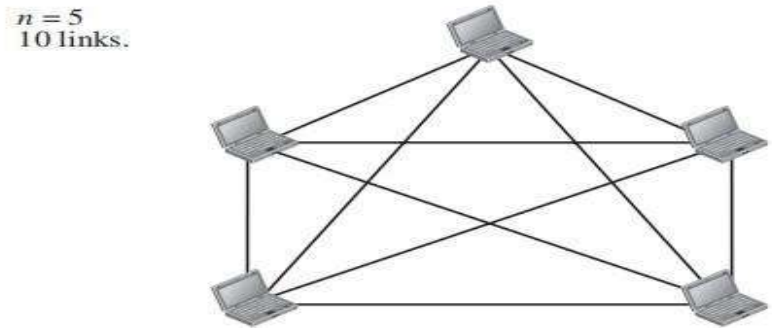
❖ Four basic topologies:

- ❖ **Mesh Topology**
- ❖ **Star**
- ❖ **Bus**
- ❖ **Ring**

NETWORKS -Physical Topology

❖ Mesh Topology

A fully connected mesh topology (five devices)



- Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes.
- Need n (n – 1) physical links.
- If each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology,
- we need n (n – 1) / 2 duplex-mode links.

➢ Find the number of **physical links in a fully connected mesh network** with n node is **n (n – 1)**

NETWORKS -Physical Topology

Advantages:

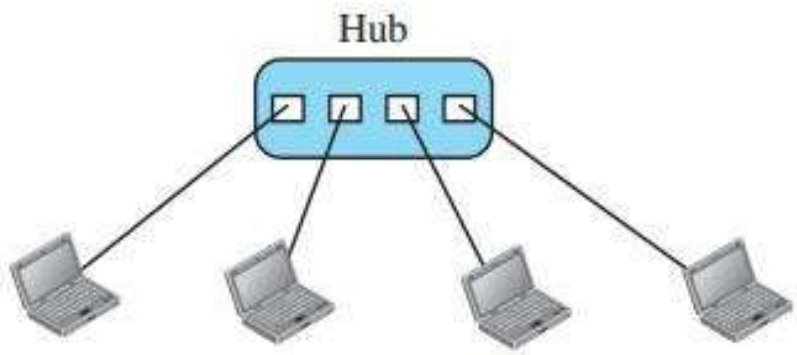
1. **Eliminating the traffic problem**
2. **Robust- A network to maintain its performance and functionality despite failures, errors, or unpredictable conditions**
3. **Privacy or security**
4. **Fault identification and fault isolation**

Disadvantages:

5. **Amount of cabling and the number of I/O ports**
6. **Installation and reconnection**
7. **Hardware required to connect each link is expensive**

NETWORKS -Physical Topology

- **Star Topology**



A hub is a basic networking device used to connect multiple computers or devices

NETWORKS -Physical Topology

- **Advantages:**
  1. **less expensive than a mesh topology, easy to install and reconfigure and less cabling needed**
  2. **If one link fails, only that link is affected**

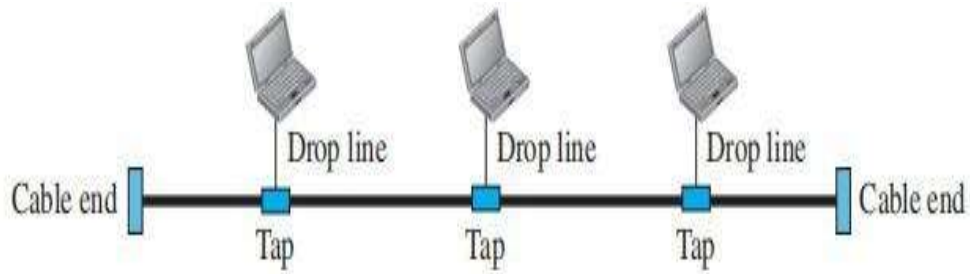
- **Disadvantage**
  1. **The dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.**

NETWORKS -Physical Topology

❖ Bus Topology

- ❖ **Long cable acts as a backbone to link all the devices in a network**

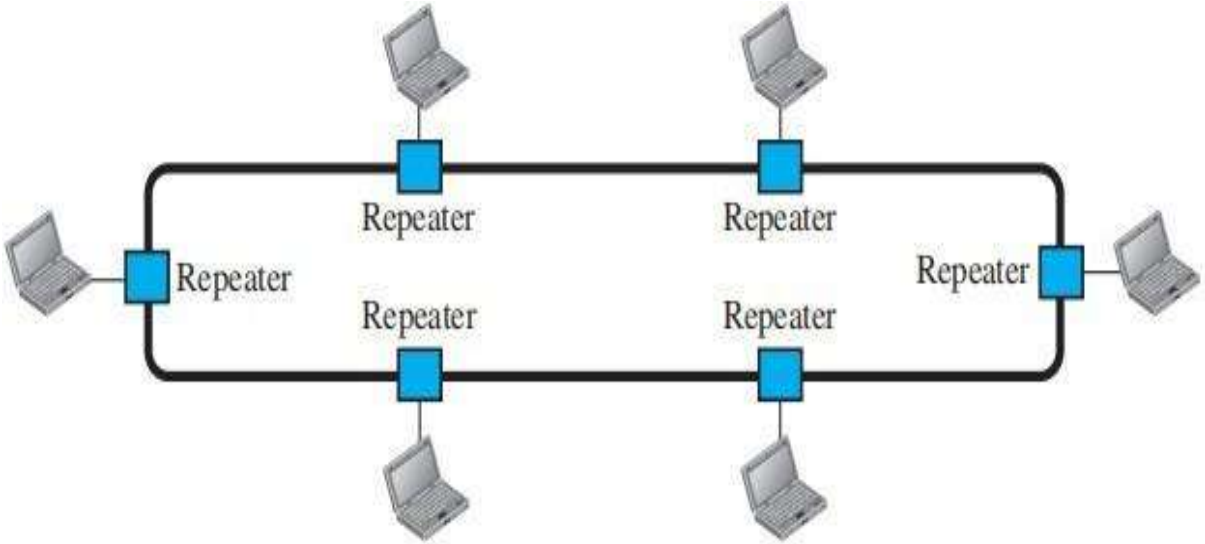
net



- **Traditional Ethernet LANs**

- **Advantages:**
    1. **Ease of installation , bus uses less cabling than mesh or star topologies**
- **Disadvantage**
    1. **Reconnection and fault isolation**
    2. **Fault or break in the bus cable stops all transmission**
    3. **The damaged area reflects signals back in the direction of origin, creating noise in both directions - cable damage leads to signal reflection because the signals are no longer properly terminated at the broken point. This reflected signal causes noise and collisions, disrupting the entire network's communication.**

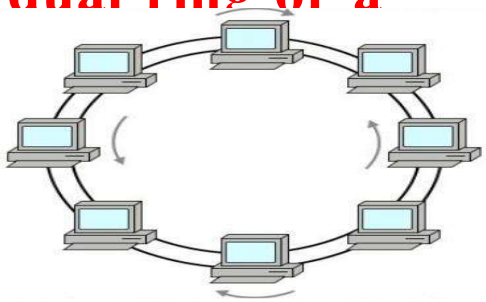
❖ Ring Topology



- **Advantages:**
  1. **A ring is relatively easy to install and reconfigure**
  2. **To add or delete a device requires changing only two connections.**

- **Disadvantage**
  1. **Unidirectional traffic**
  2. **A break in the ring (such as a disabled station) can disable the entire network**

This weakness can be solved by using a dual ring or a switch capable of closing off the break



Today's Task

Some other very famous topology: ?

Today’s Task

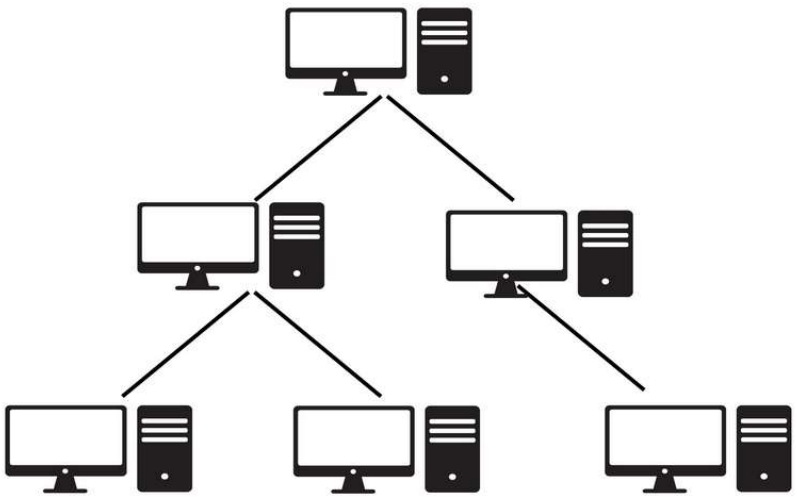
Some other very famous topology: ?

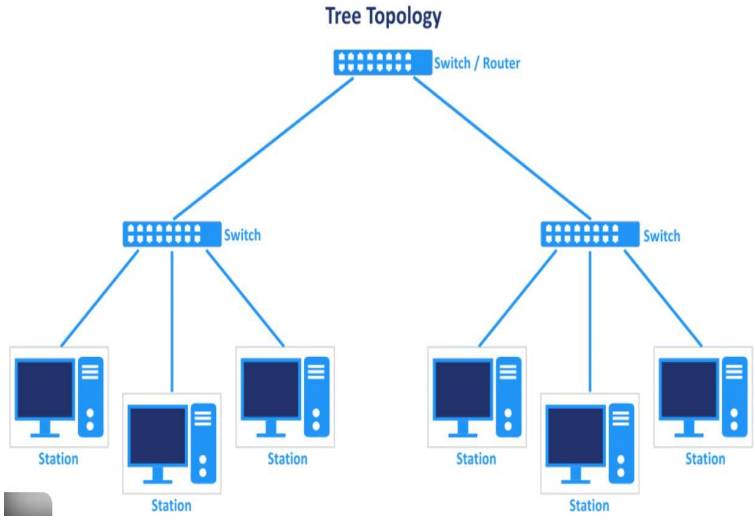
Tree Topology

Hybrid

Daisy Chain

❖ Tree Topology





Tree topology is a hierarchical network topology in which multiple star networks are connected to a main backbone cable.

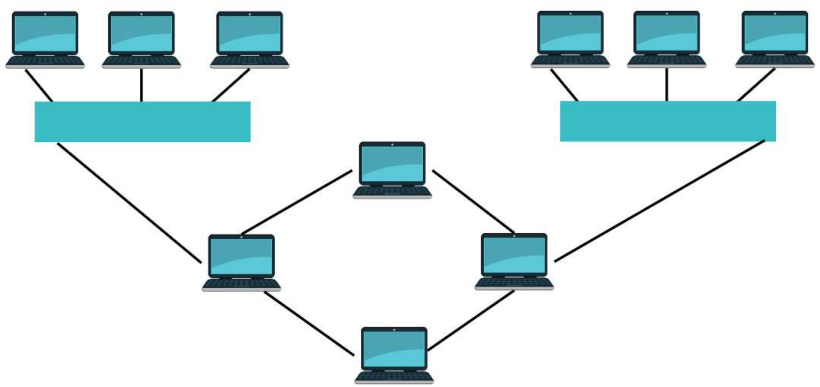
Details:
It has a root node, intermediate nodes, and leaf nodes, forming a tree-like structure. Commonly used in large organizations.

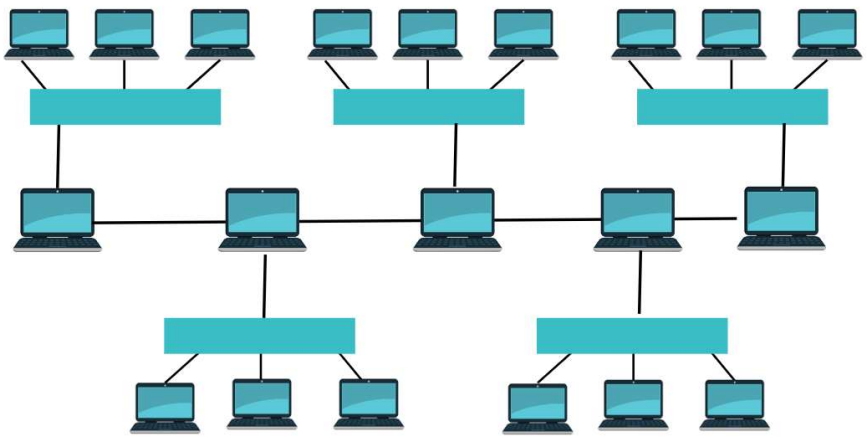
- **Advantages:**
    - **Scalability**: It is extremely easy to expand the network by adding more "branches" (star networks) to the main backbone.
    - **Fault Isolation:** If one branch (star network) fails, it does not affect the other branches or the main backbone.
    - **Organized Management:** The hierarchical structure makes it easier to manage and maintain large networks by grouping devices .

Disadvantage

- **Root Dependency:** If the main backbone cable or the root node fails, the entire network can be crippled.
- **Complexity:** Configuration and maintenance are more difficult than simpler topologies like Star or Bus.
- **Cost:** It requires a large amount of cabling and multiple hubs/switches, making it expensive to set up.

❖ Hybrid Topology





Hybrid topology is a network topology formed by combining two or more different topologies.

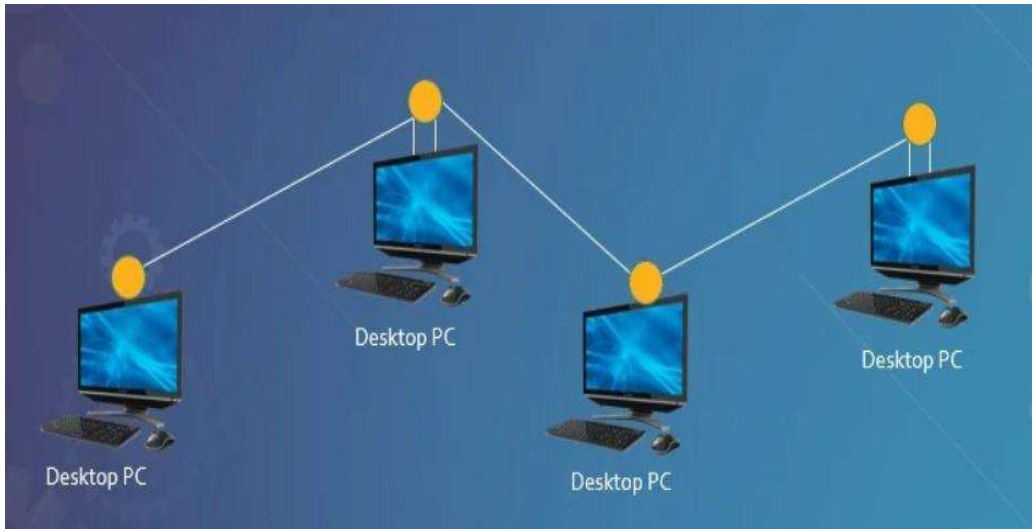
Details:
For example, star–bus or star–ring combinations are commonly used in real-world networks.

- **Advantages:**
    - **Flexibility:** It is the most flexible topology because it can be designed to meet the specific needs and constraints of an organization/user.
    - **Scalability:** You can easily add new sections using different topologies without disturbing the existing architecture.
    - **Reliability:** Fault detection and isolation are generally efficient because problems in one specific topology section often do not bring down the whole hybrid network.

Disadvantage

- **Complex Design:** Designing a hybrid network is challenging because it requires making different protocols and hardware work together.
- **Expensive:** It often requires expensive and advanced networking devices (like intelligent hubs or routers) to bridge the different topologies.
- **Maintenance:** Troubleshooting can be difficult due to the variety of configurations involved.

❖ Daisy Chain Topology



Daisy chain topology is a network topology where devices are connected sequentially in a linear chain.

Details:
Each device is connected to the next device, forming a simple chain structure.

Advantages:

- Simple design
- Less cabling required
- Easy installation

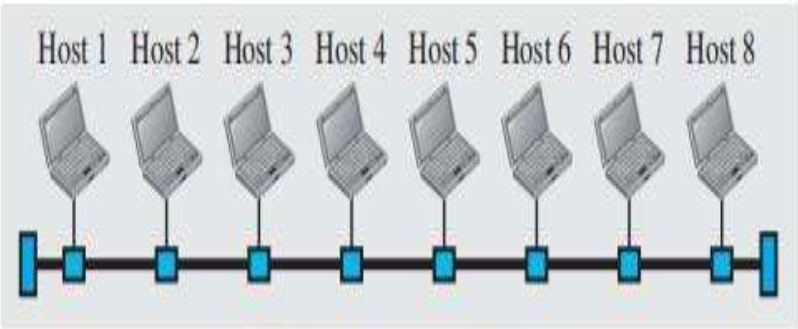
Disadvantage

- Single point of failure
- Poor scalability
- Difficult fault detection

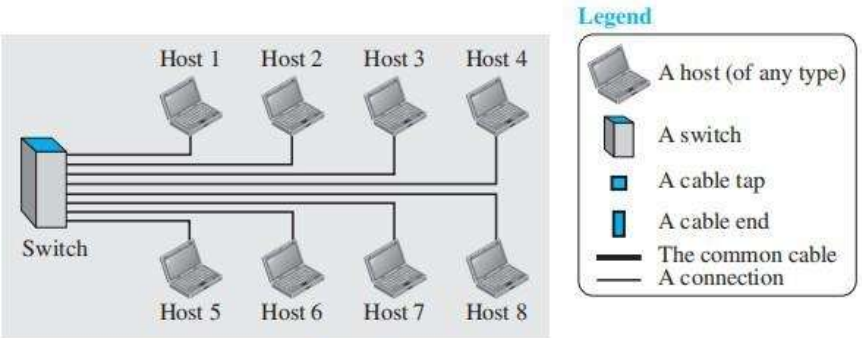
Comparison Chart of Network Topologies

Attributes	Bus	Star	Mesh	Ring	Hybrid	Tree	Daisy Chain
Network Structure	Single backbone cable	Central hub/switch	Many interconnections	Circular loop	Combination of topologies	Hierarchical (star + bus)	Sequential chain
Installation Cost	Very Low	Medium	Very High	Medium	High	Medium–High	Very Low
Reliability	Low	High	Very High	Medium	High	Medium	Very Low
Scalability	Poor	Good	Poor	Poor	Very Good	Very Good	Poor
Fault Tolerance	Very Low	Medium	Excellent	Low	High	Medium	Very Low
Performance	Low (traffic dependent)	High	Very High	Medium	High	High	Low
Ease of Maintenance	Difficult	Easy	Very Difficult	Difficult	Complex	Moderate	Difficult
Cable Requirement	Least	More than bus	Maximum	Moderate	High	High	Least
Practical Applications	Small LANs, labs	Offices, Wi-Fi networks	Internet backbone, defense	Legacy Token Ring	Corporate campuses	Universities, schools	Printers, industrial systems

Network Types- LAN (Local Area Network)



a. LAN with a common cable (past)



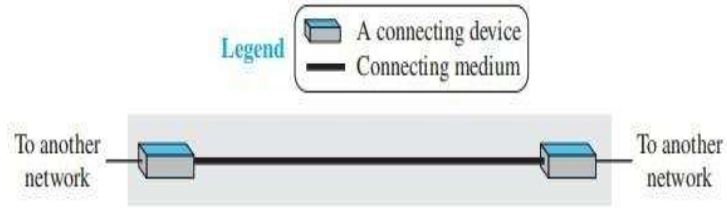
b. LAN with a switch (today)

- **In the past, all hosts in a network were connected through a common cable, which meant that a packet sent from one host to another was received by all hosts**
- **The intended recipient kept the packet; the others dropped the packet**

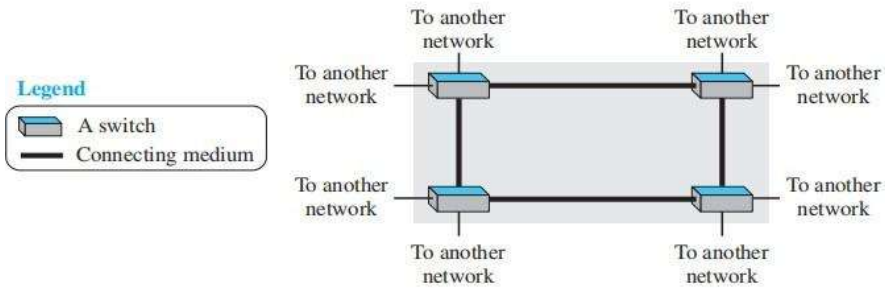
- **Most LANs use a smart connecting switch, which is able to recognize the destination address of the packet and guide the packet to its destination without sending it to all other hosts**

Network Types- WAN (Wide Area Network)

- WAN has a wide geographical span, spanning a town, a state, a country, or even the world
- WAN interconnects connecting devices such as switches, routers, or modems.
- WANs today:
  - Point-to-point WANs- Establishes a dedicated and direct connection between two devices like router,switched etc
  - Switched WANs -Uses a shared infrastructure, where multiple devices or sites are connected through switching nodes.



point-to-point WAN



Switched WAN

Network Types- Switching

- Video: https://youtu.be/G7n8thqwO2c?si=iCY9Kk0ECXcPnLAc

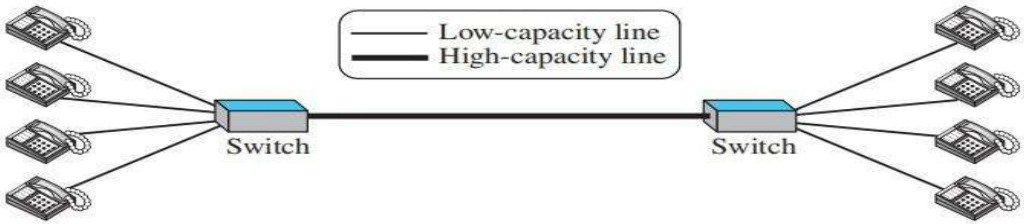
- **Switching** refers to the process of selecting the path that data will take across a network to reach its destination
- A switch needs to forward data from a network to another network when required

Two types of switched networks are,

- **Circuit-Switched Network**
- **Packet-Switched Network**

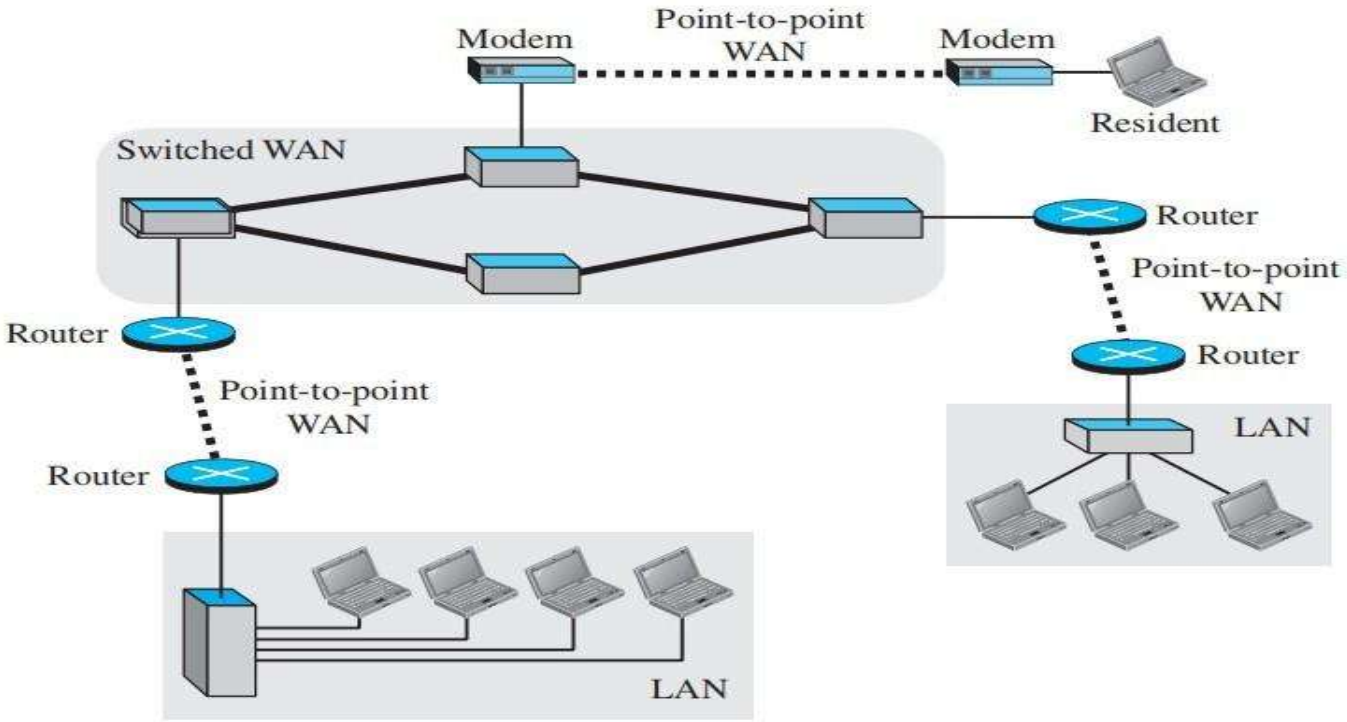
Circuit-Switched Network

- In a circuit-switched network, a dedicated connection, called a **circuit**, is always available between the two end systems; the switch can only make it active or inactive.



- **Circuit switching** was very common in **telephone networks** in the past, today we are using **packet switching** for telephone network

Circuit-Switched Network



A heterogeneous network made of four WANs and three LANs

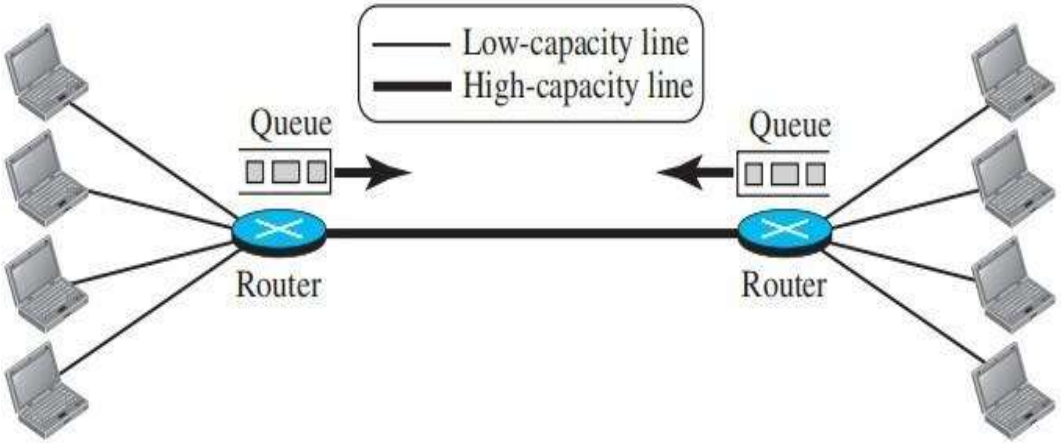
Packet-Switched Network

- In a computer network, the communication between the two ends is done in blocks of data called **packets**.

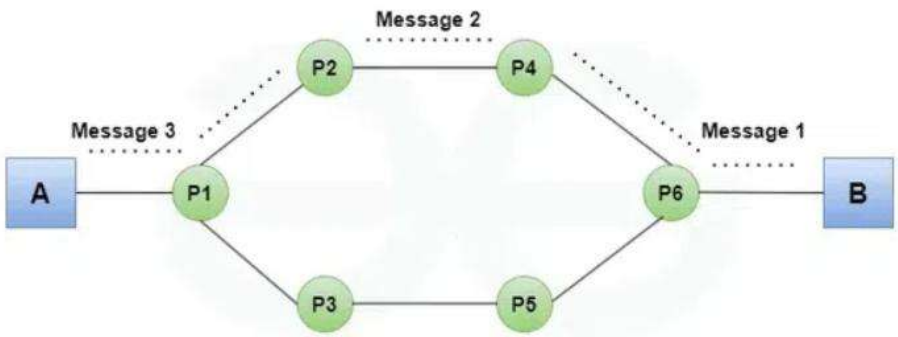
or

- Packets are **small units of data** that are transmitted over a network
- A **router** in a packet-switched network has a **queue** that can store and forward the packet.

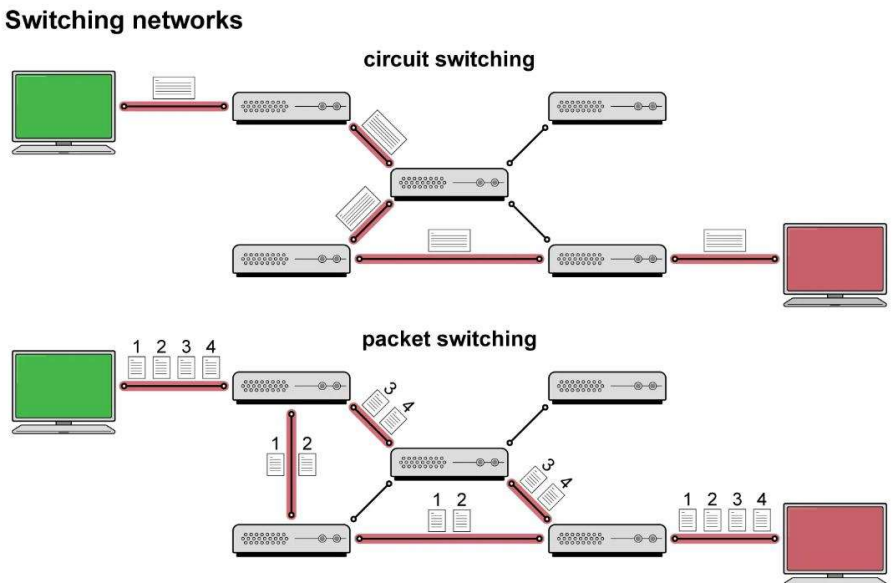
Packet-Switched Network



Circuit Switched vs Packet Switched Network



Circuit Switching



Circuit & Packet Switching

Circuit Switched vs Packet Switched Network

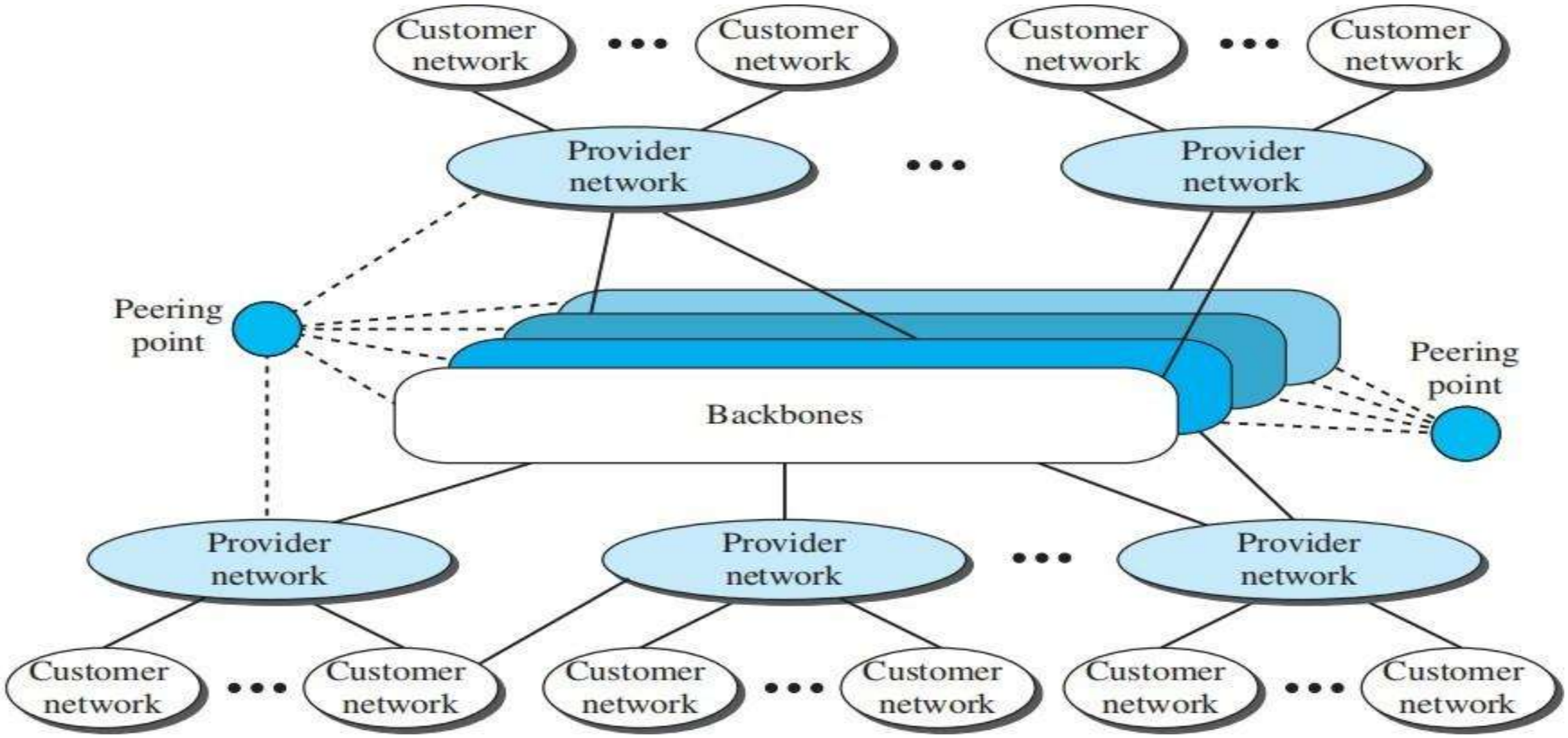
Attribute	Circuit Switched Network	Packet Switched Network
Basic Idea	A dedicated physical path is established before data transmission begins	Data is divided into packets and sent independently over shared paths
Connection Type	Connection-oriented	Connectionless (mostly)
Path Establishment	Required before communication	Not required
Data Transmission	Continuous stream of data	Data sent in small packets
Bandwidth Usage	Fixed and reserved for entire session	Shared dynamically among users
Delay	Initial setup delay, then constant delay	Variable delay (depends on traffic)
Reliability	Very reliable once connection is established	Reliability handled by protocols (e.g., TCP)
Fault Tolerance	Low (single path failure breaks connection)	High (packets can take alternate routes)
Efficiency	Low (bandwidth wasted during silence)	High (better bandwidth utilization)
Cost	Expensive	Cost-effective
Quality of Service (QoS)	Guaranteed	Not guaranteed by default
Examples of Networks	Traditional telephone network (PSTN)	Internet
Application Examples	Voice calls, leased telephone lines	Email, web browsing, video streaming, VoIP

The Internet

- **“Network of networks”**

- **This phrase highlights how the Internet is not a single network but a vast, interconnected system of smaller networks (like local area networks or LANs, wide area networks or WANs, and others) that work together to form a global communication infrastructure.**

The Internet today



The Internet today

1. **Customer Networks**:

These are individual networks belonging to **users or organizations (e.g., home users, businesses, educational institutions)**. These networks rely on **Internet service providers (ISPs)** for access to the wider Internet

2. Provider Networks: Provider networks manage traffic between **customer networks** and **core infrastructure** like Internet backbones.

3. Backbones: The backbone represents the core infrastructure of the Internet. These are **high-capacity networks** that **carry large amounts of traffic across long distances**, such as between cities or countries.

4. Peering points are locations where different networks (such as provider networks and backbone networks) connect and exchange traffic. These points help ensure that **data can flow efficiently between different networks without bottlenecks**.

Accessing the Internet

- The Internet today is an internetwork that allows any **user to become part of it**. The user, however, needs to be physically connected to an ISP. The physical connection is normally done through a point-to-point WAN.

Using Telephone Networks

- ***Dial-up service:*** The first solution is **to add to the telephone line a modem that converts data to voice.** The software installed on the computer dials the ISP and imitates making a telephone connection.
- ***DSL Service (Digital Subscriber Line):*** Since the advent of the Internet, some telephone companies have upgraded their telephone lines to provide higher speed Internet services to residences or small businesses.

Accessing the Internet(Cont…)

Using Cable Networks

- More and more residents over the last two decades have begun using cable TV services instead of antennas to receive TV broadcasting
- The cable companies have been upgrading their cable networks and connecting to the Internet. A residence or a small business can be connected to the Internet by using this service
- It provides a higher speed connection, but the speed varies depending on the number of neighbors that use the same cable

Using Wireless Networks

- With the growing wireless WAN access, a household or a small business can be connected to the Internet through a wireless WAN

Accessing the Internet(Cont…)

Direct Connection to the Internet

- A large organization or a large corporation can itself become a local ISP and be connected to the Internet
- This can be done if the organization or the corporation **leases a high-speed WAN from a carrier provider** and connects itself to a **regional ISP**
- ***For example, a large university with several campuses can create an internetwork and then connect the internetwork to the Internet***

PROTOCOL LAYERING

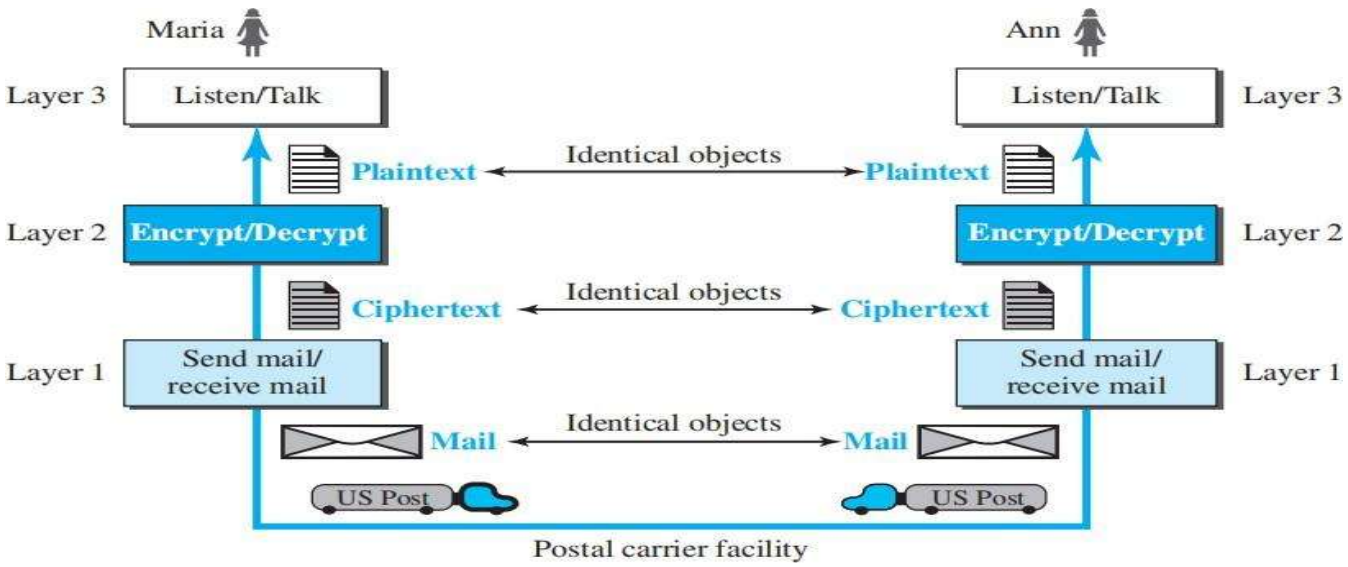
- When communication is simple, we may need only one simple protocol; when the communication is complex, we may need to divide the task between different layers, in which case we need a protocol at each layer

Scenarios:

First Scenario:

PROTOCOL LAYERING

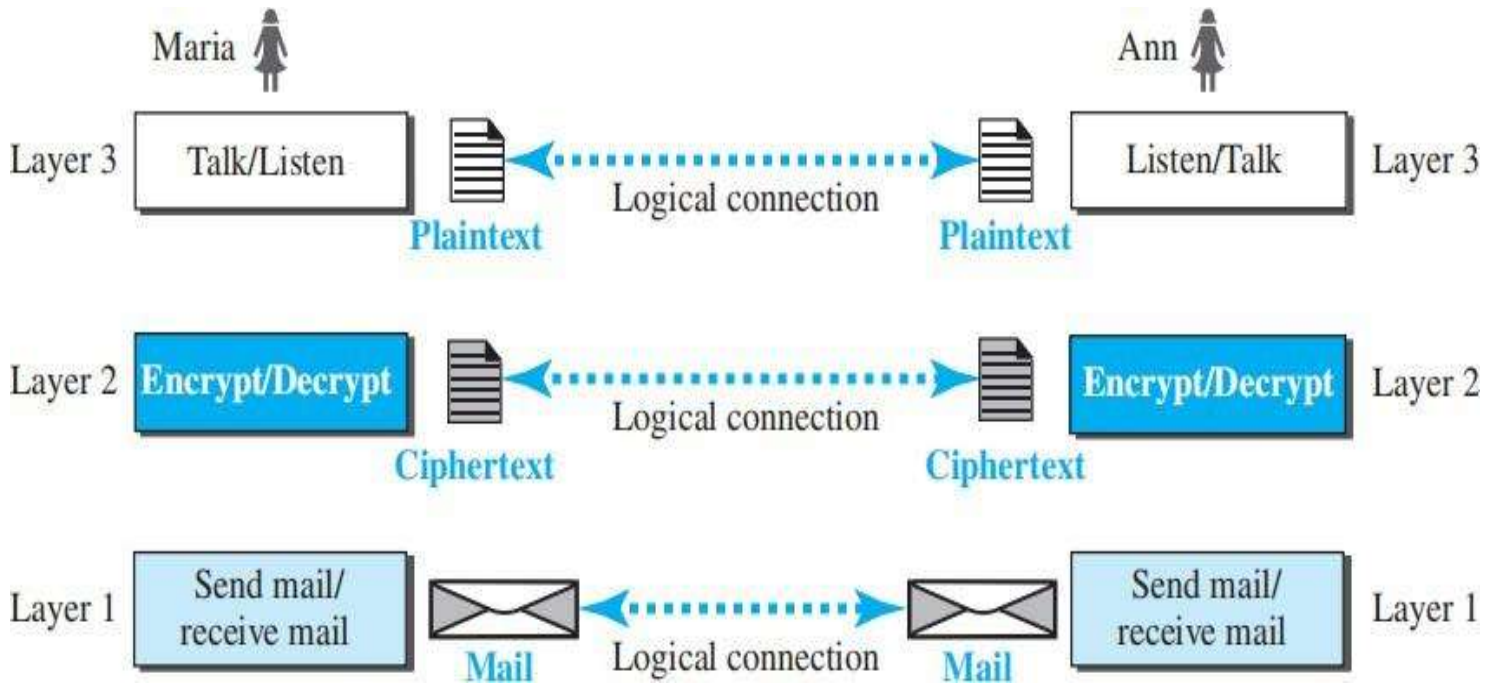
- **Second Scenario :**



Principles of Protocol Layering

- **First Principle :**
  - The first principle dictates that if we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction.
  - **Example: The third layer task is to listen (in one direction) and talk (in the other direction). The second layer needs to be able to encrypt and decrypt. The first layer needs to send and receive mail**
- **Second Principle :**
  - The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical
  - **Example : Under layer 3 at both sites should be a plaintext letter. The object under layer 2 at both sites should be a ciphertext letter. The object under layer 1 at both sites should be a piece of mail**

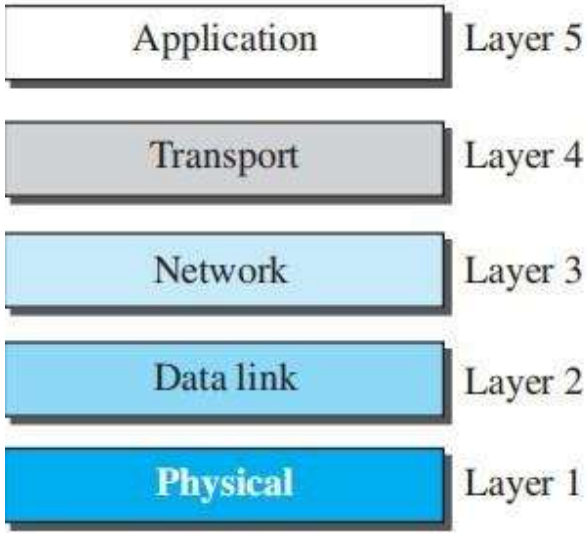
Logical Connections



TCP/IP PROTOCOL SUITE (Transmission Control Protocol/Internet Protocol)

- It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality

Layered Architecture:



Description of Each Layer

Application Layer:

- Communication at the application layer is between two processes (two programs running at this layer).
- To communicate, a process sends a request to the other process and receives a response.
- **Process-to-process** communication is the duty of the application layer.
- **Web browser (client)** communicating with a **web server (server)** is an example of process-to-process communication.

- **Protocols:**
  - **Hypertext Transfer Protocol (HTTP) -** Accessing the World Wide Web (WWW)
  - **Simple Mail Transfer Protocol (SMTP) -** Electronic mail (e-mail) service
  - **File Transfer Protocol (FTP) -** Used for transferring files from one host to another
  - **Terminal Network (TELNET) and Secure Shell (SSH) -** Accessing a site remotely
  - **Simple Network Man_x0002_agement Protocol (SNMP) -** Used by an administrator to manage the Internet at global and local levels
  - **Domain Name System (DNS) :** Used by other protocols to find the network-layer address of a computer
  - **Internet Group Management Protocol (IGMP) :**Used to collect membership in a group

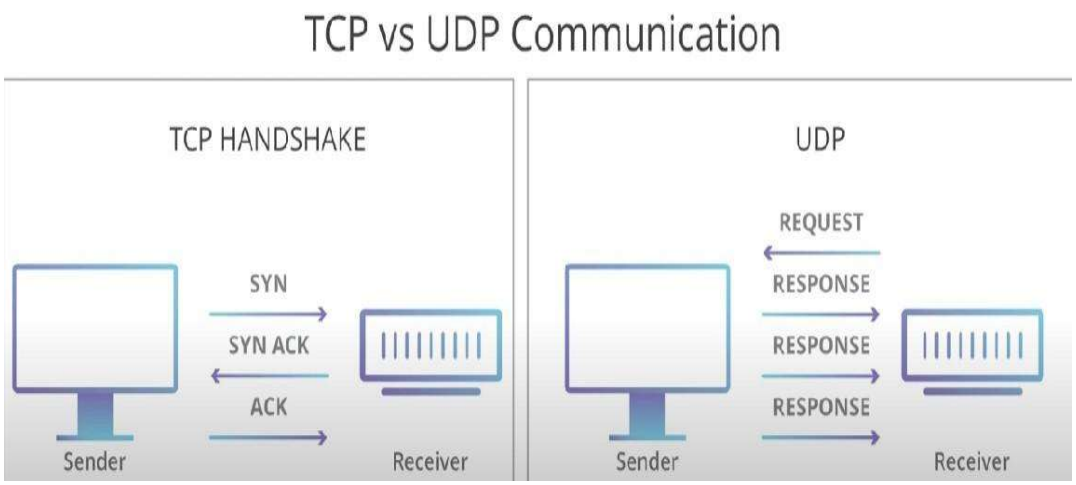
Description of Each Layer

- **Transport Layer:Data-segment** or a **user datagram**
- The transport layer at the source host gets the message from the application layer, encapsulates it in a transport layer packet and send

Application Protocol	Transport Protocol	Port Number	Description
HTTP	TCP	80	Used by web browsers and web servers
Telnet	TCP	23	Used for terminal emulation
SSH	TCP	22	Used for secure terminal emulation
FTP	TCP	20, 21	Used for file transfer
DNS	UDP	53	Used for name-to-IP resolution
SMTP	TCP	25	Used to send Email
POP3	TCP	110	Used to receive Email

Description of Each Layer

- **Protocol:**
  - **Transmission Control Protocol (TCP) - Connection-oriented protocol** that first establishes a logical connection between transport layers at two hosts before transferring data.
  - It is a **Reliable communication**
  - TCP provides :
    - Flow control
    - Error Control
    - congestion control



Description of Each Layer

- **Protocol:**
  - **User Datagram Protocol (UDP) :**
    - Connectionless protocol that transmits user datagrams without first creating a logical connection
    - UDP does not provide
      - flow, error, or congestion control
  - **Stream Control Transmission Protocol (SCTP)**
    - Designed to respond to new applications that are emerging in the multimedia

Description of Each Layer

Network Layer: (Finding Optimal Path)-Data:Fragments if the size of the packet is very large

Creating a connection between the **source computer and the destination computer**

- Communication at the network layer is **host-to-host**
- Several routers from the source to the destination, the routers in the path are responsible for choosing the **best route for each packet**

- **Protocols :**
  - **Internet Protocol (IP) : IP is used to "address" each device in that network**
    - **IP is also responsible for routing a packet from its source to its destination**
  - **Internet Control Message Protocol (ICMP)** : Helps IP to report some problems when routing a packet
  - **Internet Group Management Protocol (IGMP) :** Helps IP in multitasking

Description of Each Layer

- **Protocols :**
  - **Dynamic Host Configuration Protocol (DHCP) :** Helps IP to get the network-layer address for a host
  - **Address Resolution Protocol (ARP):** Helps IP to find the MAC Address(Media Access Control) of a host or a router

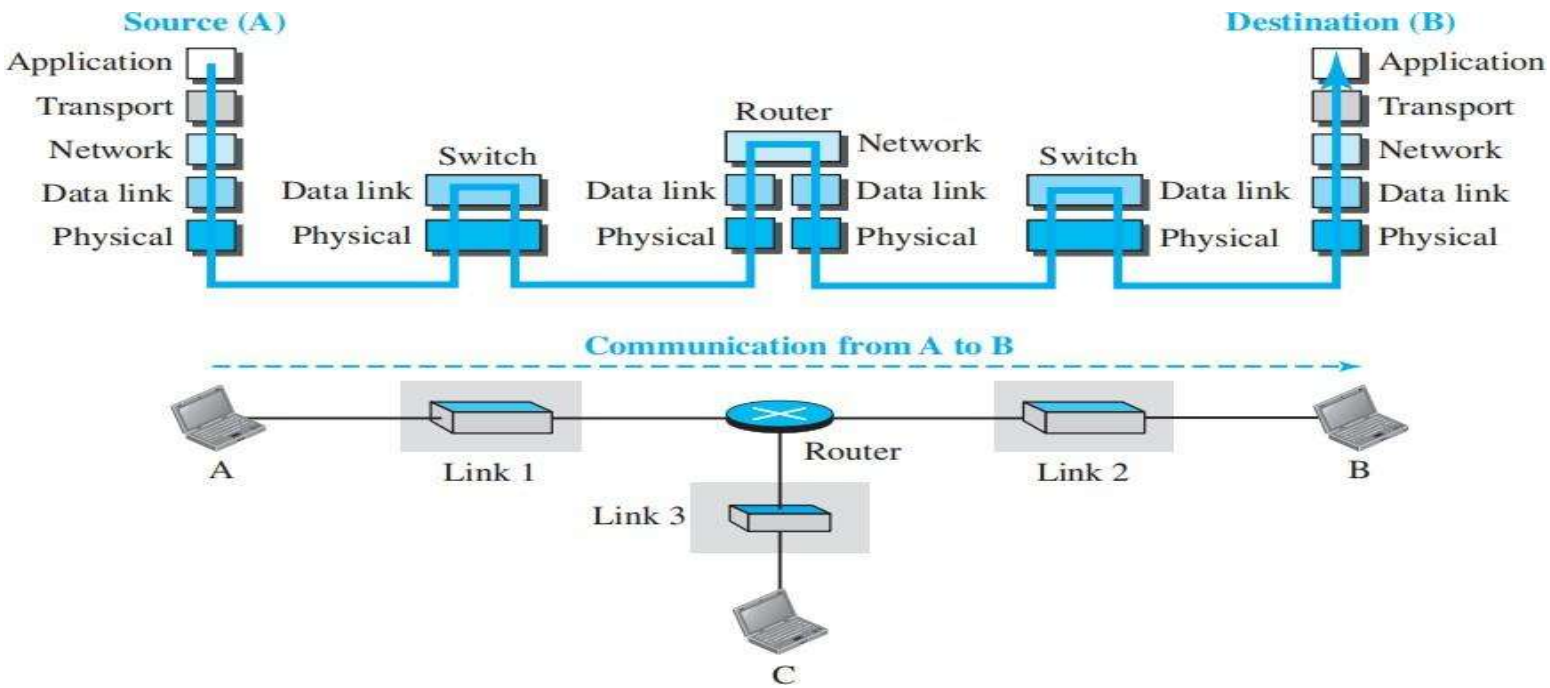
Data-link Layer:

- The data-link layer takes a **datagram and encapsulates it in a packet** called a frame
  1. A header (which includes the source and destination MAC addresses)
  2. The payload (the fragment of the datagram)
  3. A trailer (which often includes error-checking data like CRC).
- TCP/IP does not define any specific protocol for the data-link layer.

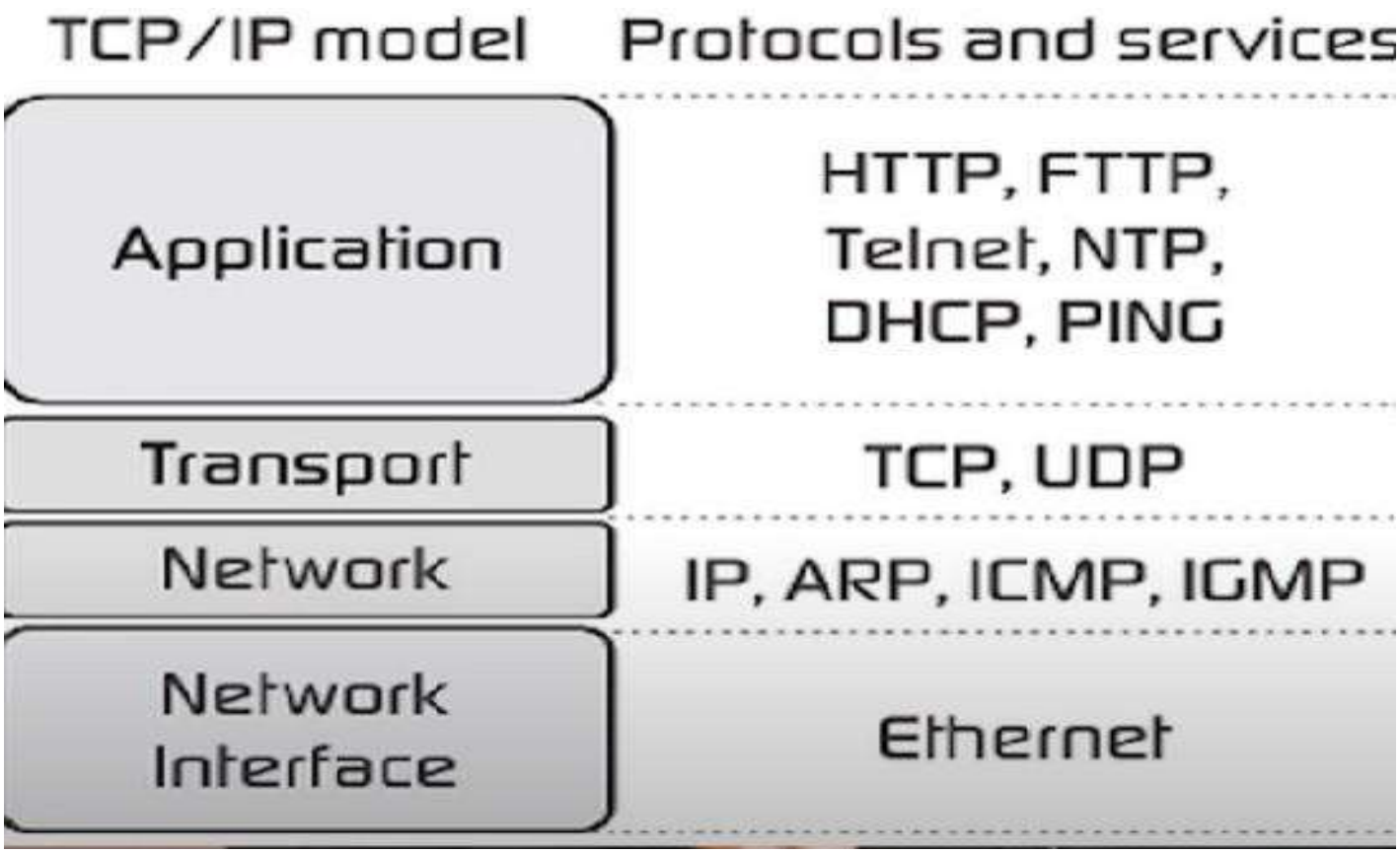
Description of Each Layer

- **Physical Layer:**
  - Physical layer is responsible for carrying individual bits in a frame across the link
  - Bits received in a frame from the data-link layer are trans_x0002_formed and sent through the transmission media

Description of Each Layer

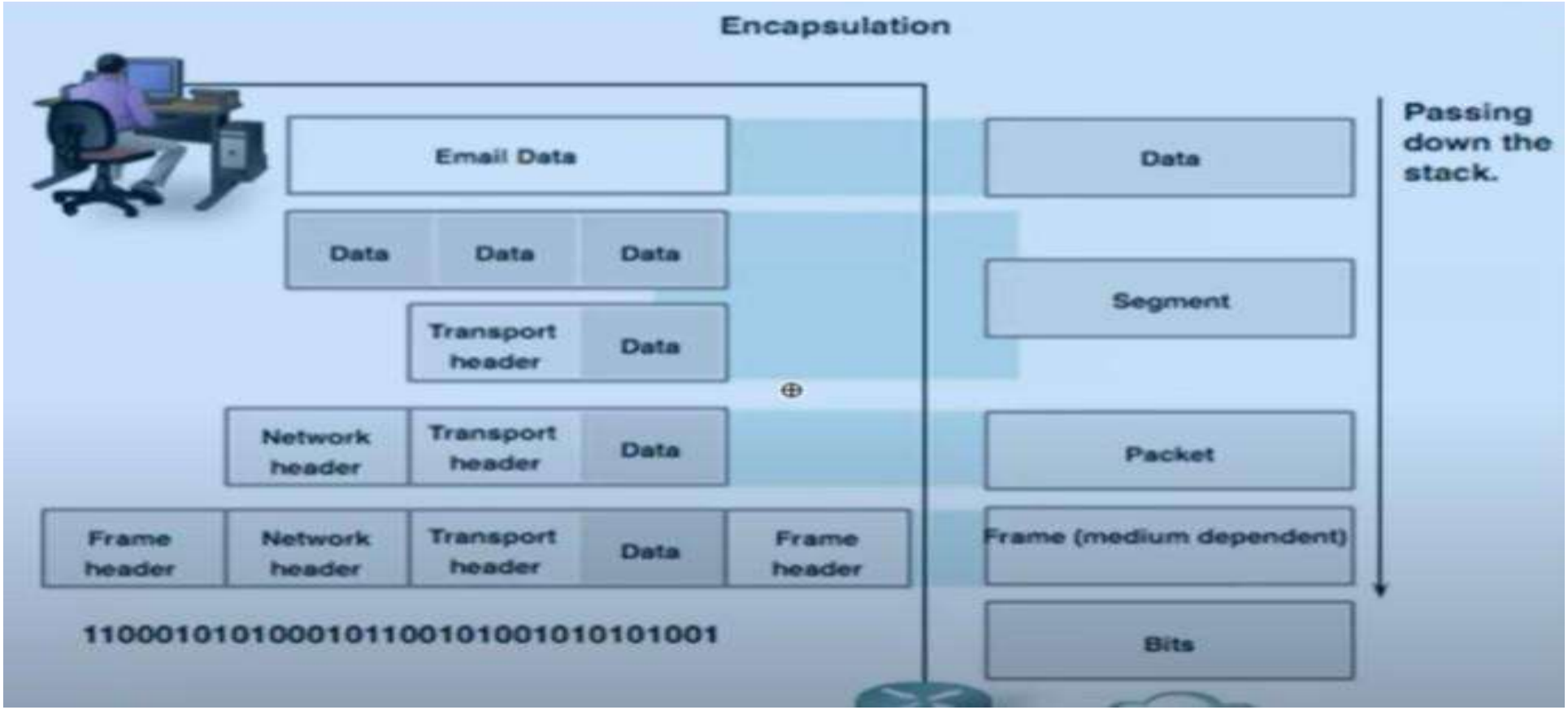


Description of Each Layer

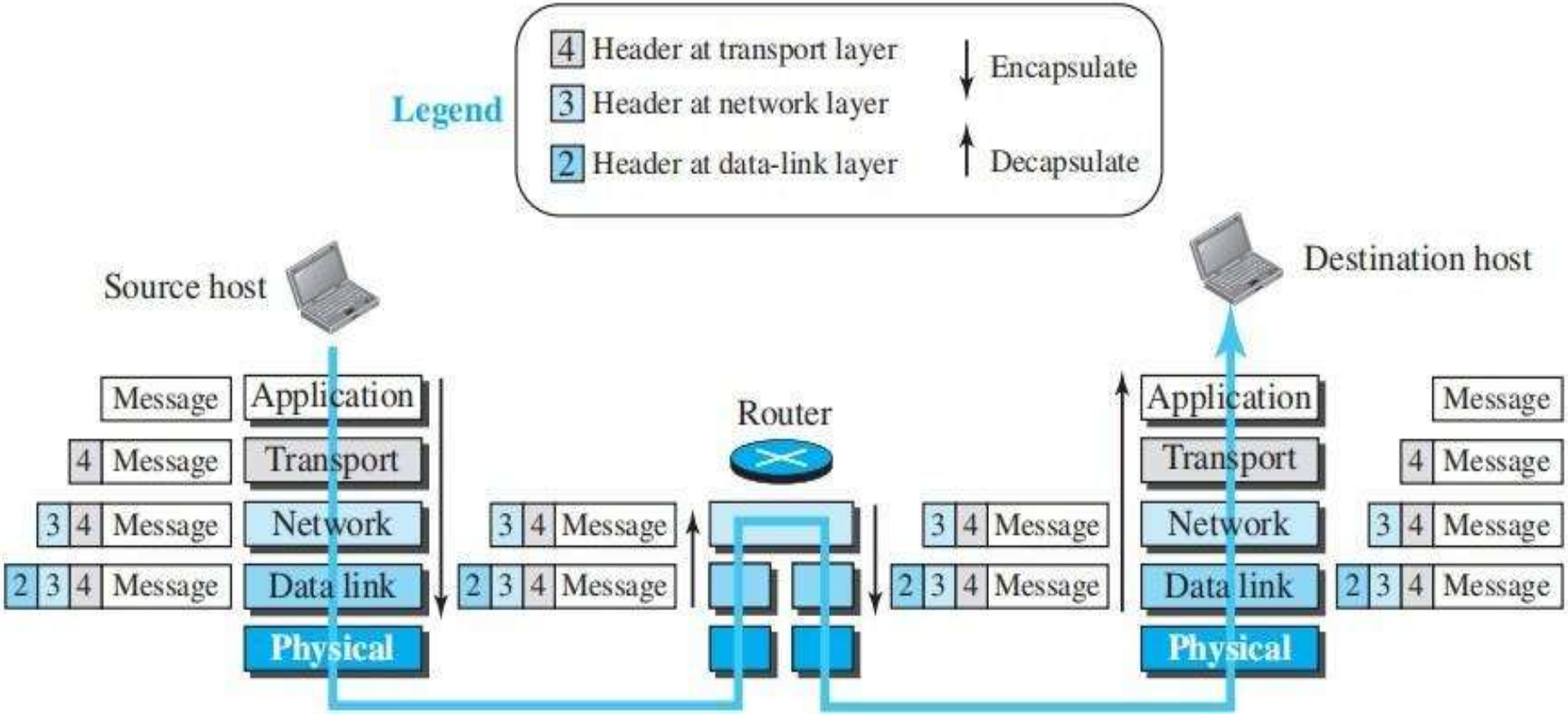


Description of Each Layer

Description of Each Layer



Encapsulation and Decapsulation



Encapsulation and Decapsulation

- **Encapsulation at the Source Host:**

1. At the **application layer**, the data to be exchanged is referred to as a message. A message normally does not contain any header or trailer, but if it does, we refer to the whole as the message. The message is passed to the transport layer

2. The **transport layer** takes the message as the payload, the load that the transportlayer should take care of. It adds the transport layer **header to the payload**, which contains the **identifiers of the source and destination** application programs that want to communicate plus some more information that is needed for the end-to-end delivery of the message, such as information needed for **flow, error control, or congestion control**. The result is the transport-layer packet, which is called the **segment (in TCP) and the user datagram (in UDP)**. The transport layer then passes the packet to the network layer.

Encapsulation and Decapsulation

- **Encapsulation at the Source Host:**

3. The network layer takes the transport-layer packet as data or payload and **adds its own header to the payload.** The **header contains the addresses of the source and destination hosts and some more information used for error checking of the header, fragmentation information, and so on**. The result is the network-layer packet, called a datagram. The network layer then passes the packet to the data-link layer.
4. The data-link layer takes the network-layer packet as data or payload and **adds its own header, which contains the link-layer addresses of the host or the next hop (the router).** The result is the link-layer packet, which is called a frame. The frame is passed to the physical layer for transmission.

Encapsulation and Decapsulation

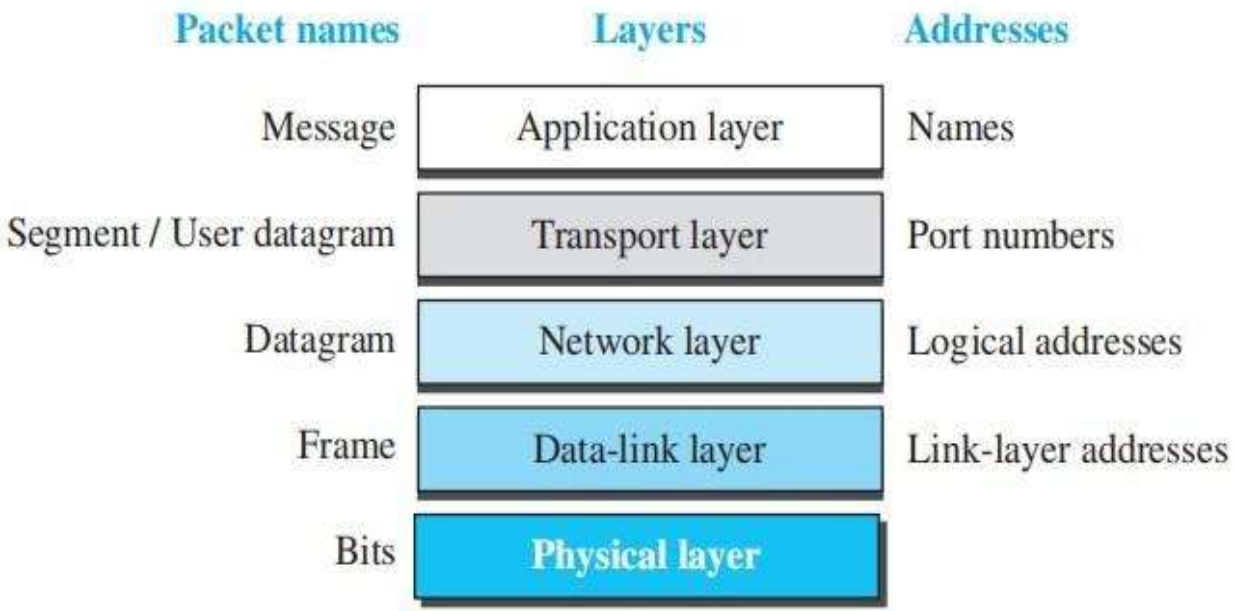
- **Decapsulation and Encapsulation at the Router**
  1. After the set of bits are delivered to the data-link layer, this layer decapsulates the datagram from the frame and passes it to the network layer
  2. The network layer only inspects the source and destination addresses in the datagram header and consults its forwarding table to find the next hop to which the datagram is to be delivered. The contents of the datagram should not be changed by the network layer in the router unless there is a need to fragment the datagram if it is too big to be passed through the next link. The datagram is then passed to the data-link layer of the next link
  3. The data-link layer of the next link encapsulates the datagram in a frame and passes it to the physical layer for transmission.

Encapsulation and Decapsulation

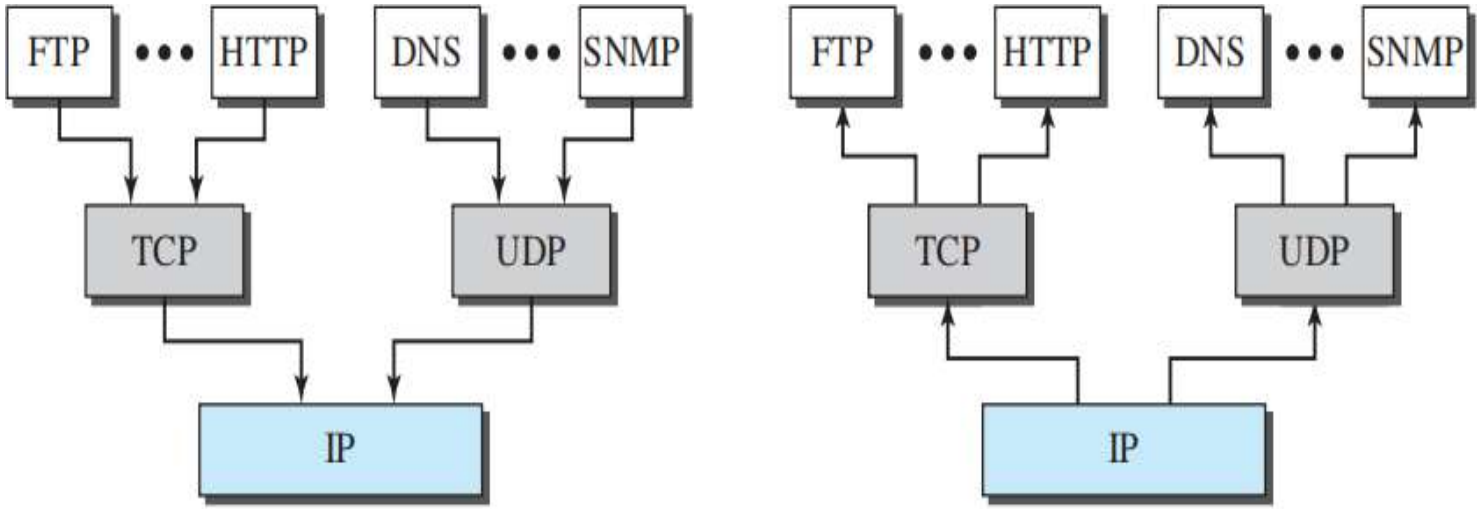
- **Decapsulation at the Destination Host:**

1. Each layer only decapsulates the packet received, removes the payload, and delivers the payload to the next-higher layer protocol until the message reaches the application layer

Addressing


Packet names
Layers
Addresses
Message
Application layer
Names
Segment / User datagram
Transport layer
Port numbers
Datagram
Network layer
Logical addresses
Frame
Data-link layer
Link-layer addresses
Bits
Physical layer

Multiplexing and Demultiplexing



a. Multiplexing at source

b. Demultiplexing at destination

Key Points of TCP/IP Model

- TCP/IP stands for Transmission Control Protocol / Internet Protocol
- It is the standard protocol suite of the Internet
- Developed by DARPA (USA)
- Used for end-to-end data communication
- It follows a 4-layer architecture
- Practical and implementation-oriented model
- Used in LAN, WAN, and the Internet

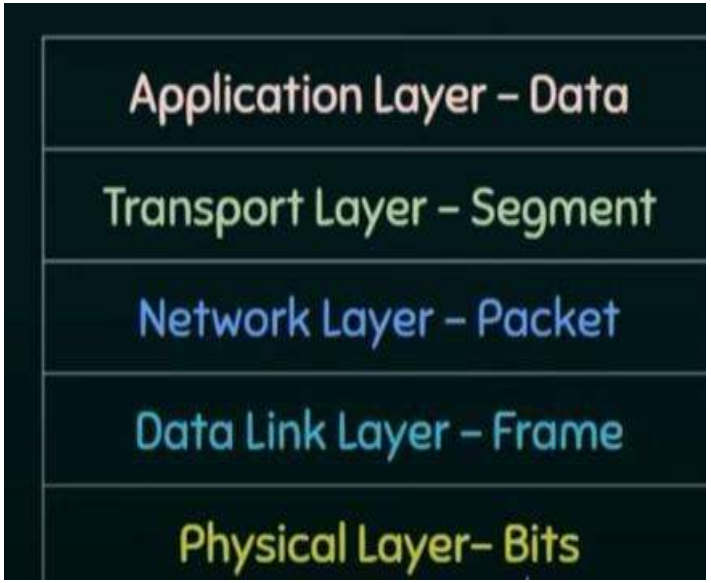
Layers of TCP/IP Model

Application Layer – HTTP, FTP, SMTP, DNS

Transport Layer – TCP, UDP

Internet Layer – IP, ICMP, ARP

Network Access Layer – Ethernet, Wi-Fi



Advantages of TCP/IP Model

- Widely used and accepted worldwide
- Highly scalable (supports small to very large networks)
- Platform independent
- Supports internetworking (connects different networks)
- Reliable communication using TCP
- Flexible (works with many underlying network technologies)

Disadvantages of TCP/IP Model

- No clear separation of services, interfaces, and protocols
- Transport layer does not guarantee delivery (UDP is unreliable)
- Not suitable for small networks
- Hard to replace protocols once standardized
- Less conceptual clarity compared to OSI model

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Today’s Task

Is Voice over IP (VoIP) using the TCP/IP model?

Today's Task

Is Voice over IP (VoIP) using the TCP/IP model? ----------→YES

Voice over IP (VoIP) works entirely on the TCP/IP model

How VoIP uses TCP/IP

Application Layer: SIP, RTP, RTCP

Transport Layer: UDP (mostly)

Internet Layer: IP

Network Access Layer: Ethernet / Wi-Fi

📌 **Why UDP is used?**

Voice needs **low delay**

Small packet loss is acceptable

TCP retransmission causes delay → poor call quality

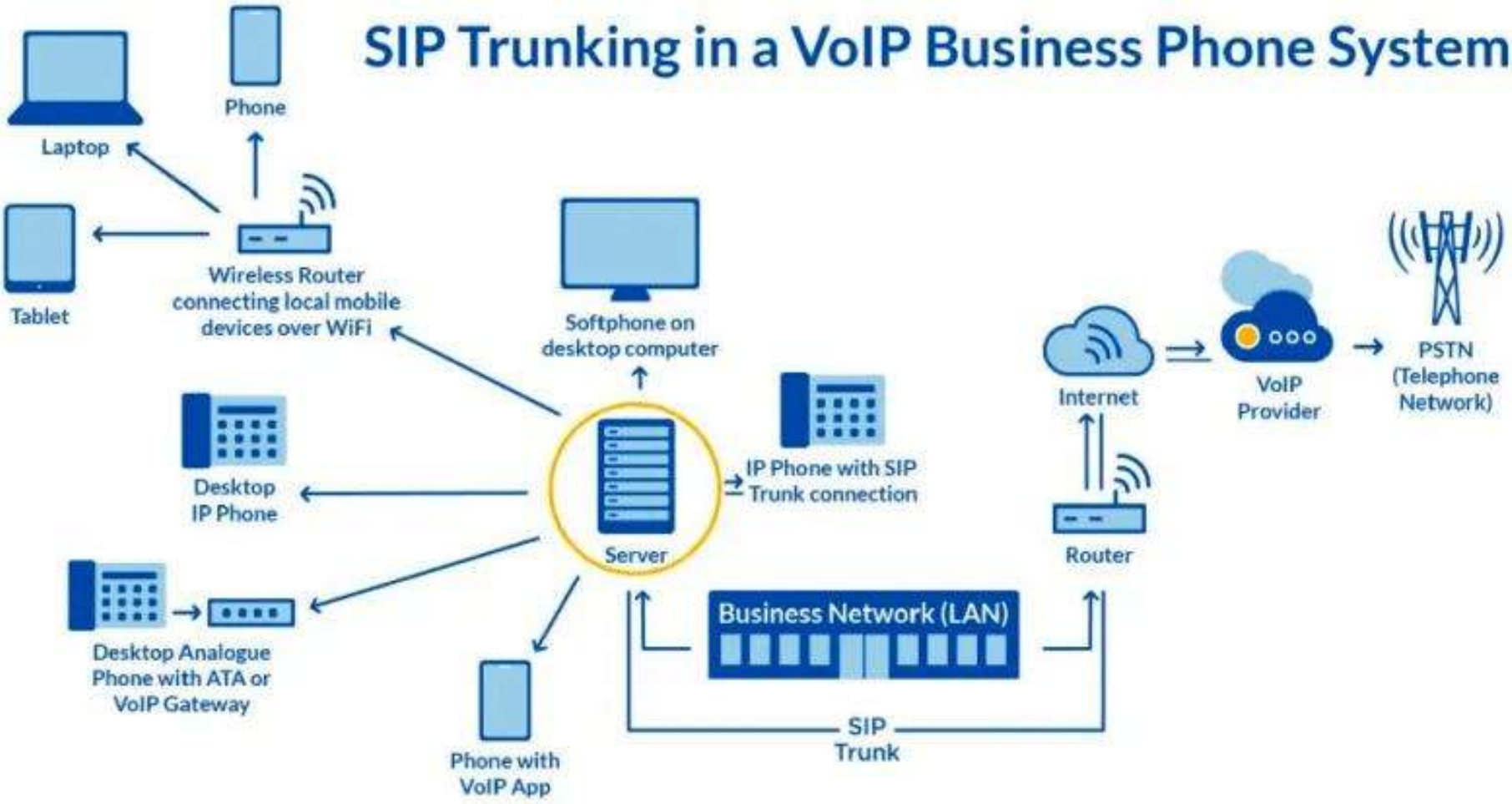
Examples of VoIP services:

Internet calls

Video conferencing

IP telephony

SIP Trunking in a VoIP Business Phone System



Today’s Task

Is WhatsApp messaging using the TCP/IP model?

Today's Task

Is WhatsApp messaging using the TCP/IP model? **YES**

WhatsApp messaging also uses the TCP/IP model.

How WhatsApp uses TCP/IP

Application Layer: WhatsApp protocol

Transport Layer: TCP (for messages)

Internet Layer: IP

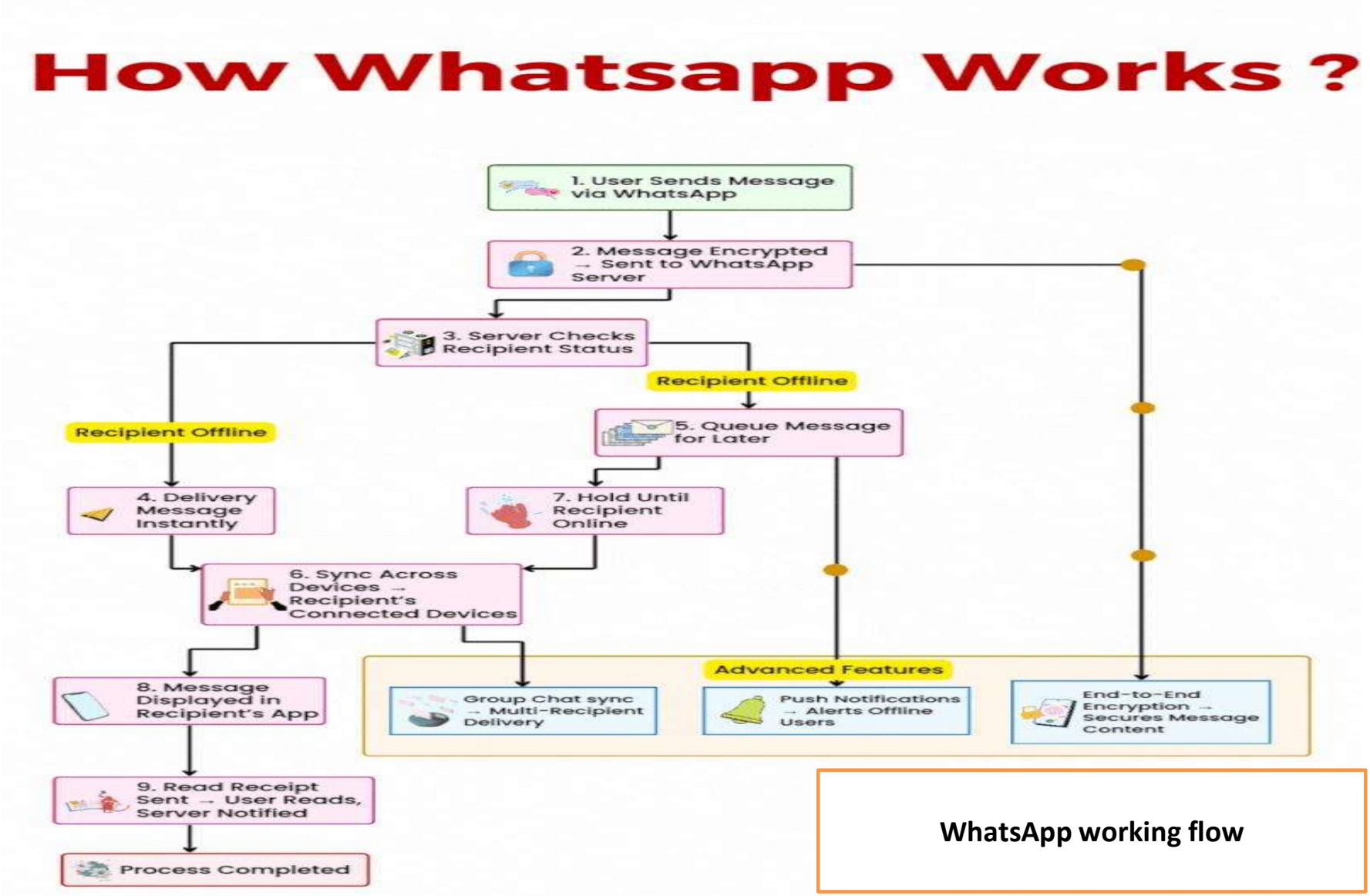
Network Access Layer: Mobile data / Wi-Fi

📌 **Why TCP is used for messaging?**

Messages must be delivered correctly

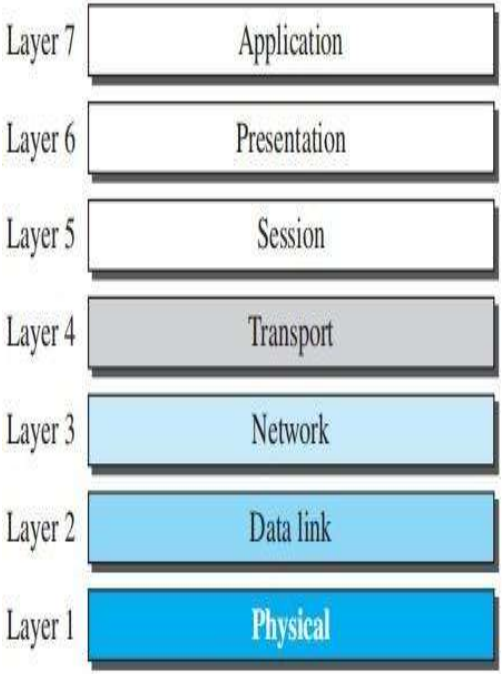
No data loss allowed

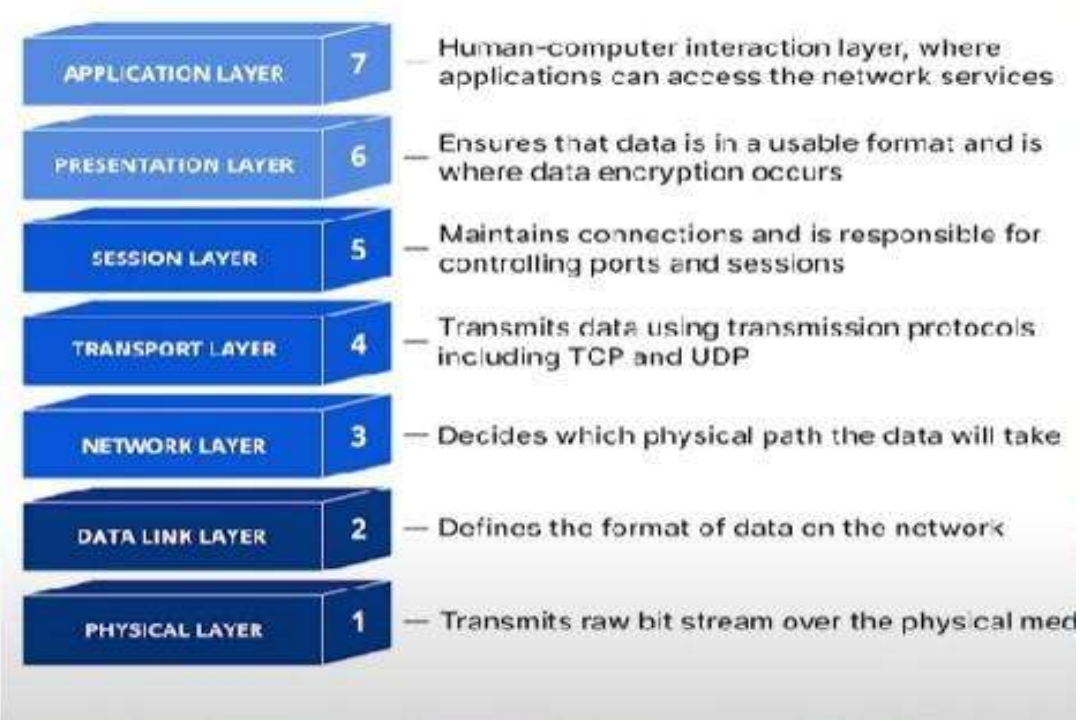
Order of messages is important



WhatsApp working flow

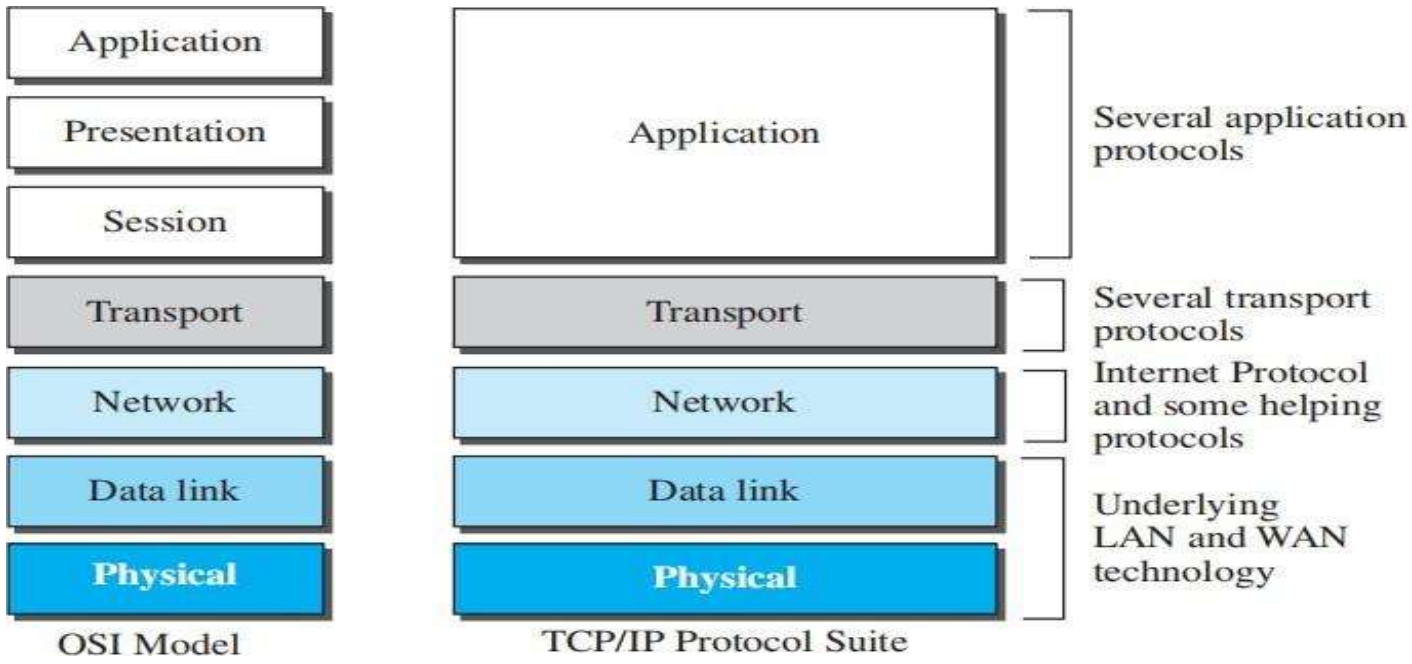
THE OSI MODEL (Open Systems Interconnection)





International Organization for Standardization (ISO)

THE OSI MODEL (Open Systems Interconnection)



THE OSI MODEL (Open Systems Interconnection)

- **Presentation:** Translation,Compression and Encryption
- **Session layer:**Authentication and security and session management

Lack of OSI Model's Success

1. OSI was completed when TCP/IP was fully in place and a lot of time and money had been spent on the suite; changing it would cost a lot.
2. Some layers in the OSI model were never fully defined.

For example, although the services provided by the presentation and the session layers were listed in the document, actual protocols for these two layers were not fully defined, nor were they fully described, and the corresponding software was not fully developed.

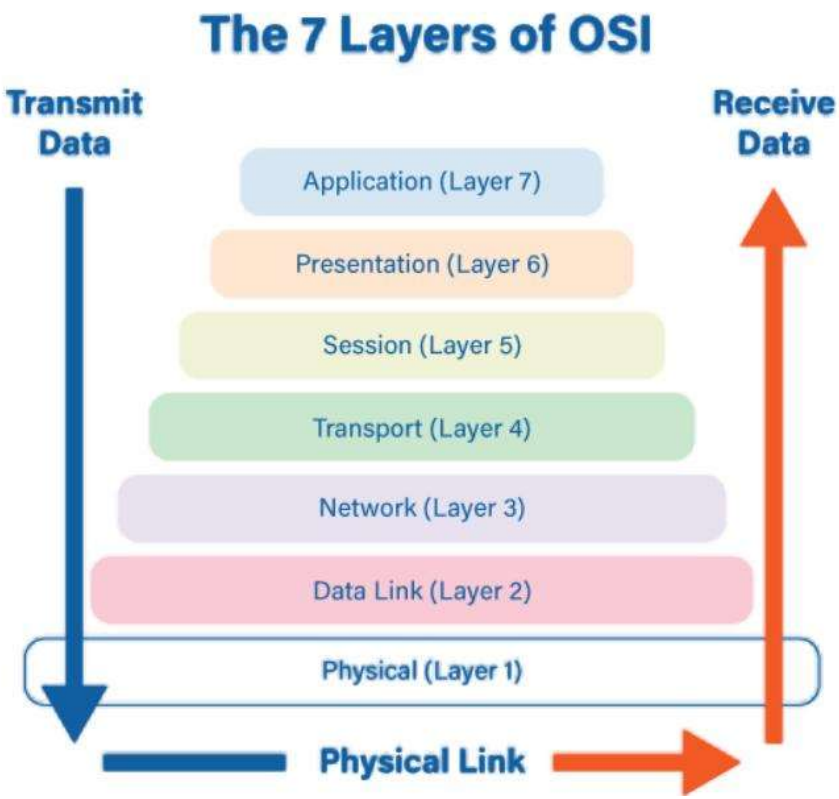
3. when **OSI was implemented by an organization** in a different application, it **did not show a high enough level of performance** to entice the Internet authority to switch from the TCP/IP protocol suite to the OSI model.

Key Points of OSI Model

- OSI stands for Open Systems Interconnection
- Developed by ISO (International Organization for Standardization)
- It is a 7-layer reference model
- Provides a standard framework for network communication
- Each layer has a specific function
- Used mainly for learning, design, and troubleshooting
- OSI is a theoretical model, not a protocol suite

Seven Layers of OSI Model

- Application
- Presentation
- Session
- Transport
- Network
- Data Link
- Physical



Advantages of OSI Model

- Clear layer separation (easy to understand & design)
- Standard reference model for networking
- Interoperability between different vendors
- Simplifies troubleshooting
- Layer independence (changes in one layer don’t affect others)
- Supports modular development

Disadvantages of OSI Model

- Theoretical model (not directly implemented)
- Complex structure with 7 layers
- Session and Presentation layers rarely used separately
- Slower adoption compared to TCP/IP
- Higher overhead due to extra layers

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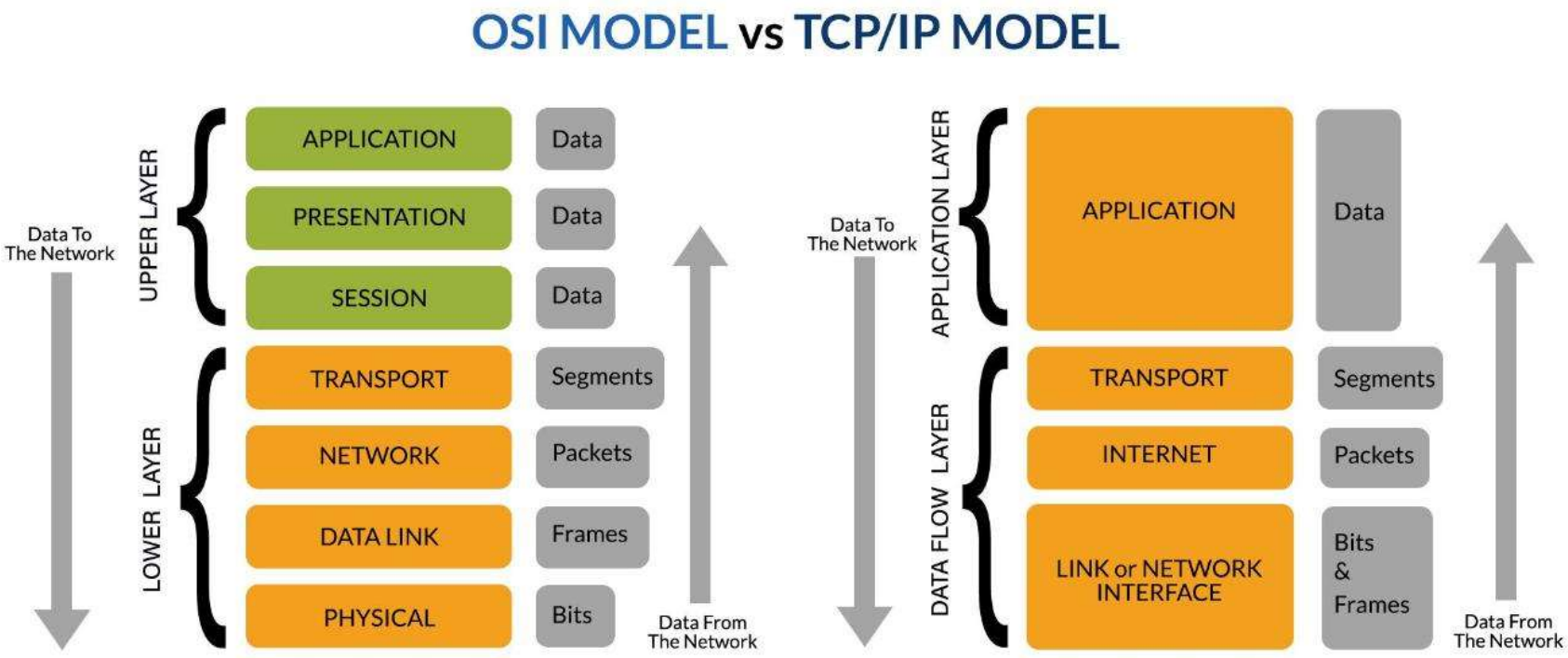
OSI vs TCP/IP Comparison Table (Very Important for Exams)

Basis	**OSI Model**	**TCP/IP Model**
Full Form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Developed By	ISO	DARPA (USA)
Number of Layers	**7 Layers**	**4/5 Layers**
Nature	Theoretical / Reference model	Practical / Implementation model
Layer Separation	Very clear	Less clear
Session & Presentation Layers	Present	Not present
Transport Layer	Supports connection-oriented & connectionless	TCP (reliable) & UDP (unreliable)
Network Layer	Supports connection-oriented & connectionless	Supports only connectionless (IP)
Used In Practice	❌ Rare	✅ Widely used (Internet)
Ease of Understanding	Easy to understand	Slightly complex
Example Use	Teaching & troubleshooting	Real-world networking

OSI vs TCP/IP Comparison Table (Very Important for Exams)



OSI vs TCP/IP – Practical Use Comparison

Aspect	**OSI Model**	**TCP/IP Model**
Nature	Reference / Conceptual	Practical / Implemented
Used In	Education, design, troubleshooting	Internet, LAN, WAN
Layers	7	4
Real-time implementation	No	Yes
Main Purpose	Understanding & standardization	Data communication